



BACHELOR'S THESIS IN ENGINEERING PHYSICS

UNIVERSITAT POLITÈCNICA DE CATALUNYA
ESCOLA TÈCNICA SUPERIOR D'ENGINYERIA DE TELECOMUNICACIÓ DE
BARCELONA

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On-chip polarization-entangled photon-pair source based on silicon photonics integration

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September 2026

Abstract

We present the experimental characterization of a novel monolithically integrated polarization-entangled photon-pair source fabricated on a silicon-on-insulator (SOI) platform. The device combines a polarization splitter-rotator (PSR) with a common-path Sagnac loop, in which two counter-propagating spontaneous four-wave mixing (SFWM) processes are coherently combined to generate polarization entanglement, without any off-chip polarization control or thermal stabilization. Classical characterization of the fabricated chip confirmed broadband stimulated four-wave mixing generation over a bandwidth of approximately 1480 nm to 1625 nm, encompassing the telecom C-band.

Entanglement certification was carried out through two-photon polarization interference, yielding fringe visibilities of 84.13% in the rectilinear basis and 88.65% in the diagonal basis, both exceeding the 70.71% threshold required for a CHSH inequality violation. The CHSH S parameter was experimentally evaluated at $S = 2.41 \pm 0.008$, a violation of the classical bound by 51.9 standard deviations. These results certify that the source generates genuine polarization-entangled photon pairs. Being robust and passively stable by construction, the source poses as a promising for future satellite-based quantum key distribution schemes.

Català

Es presenta la caracterització experimental d'una nova font de parells de fotons entrellaçats en polarització, monolíticament integrada i fabricada sobre una plataforma de silici sobre aïllant (SOI). El dispositiu combina un separador-rotador de polarització (PSR) amb un bucle de Sagnac de camí comú, en el qual dos processos de mescla espontània de quatre ones (SFWM) contrapropagants es combinen coherentment per generar entrellaçament en polarització, sense necessitat de cap control de polarització fora del xip ni d'estabilització tèrmica. La caracterització clàssica del xip fabricat permet verificar la generació de mescla estimulada de quatre ones en una banda ampla, aproximadament entre 1480 nm i 1625 nm, que inclou la banda C de les telecomunicacions.

La certificació de l'entrellaçament s'ha dut a terme mitjançant interferència de polarització de dos fotons, obtenint visibilitats de les franges del 84,13% en la base rectilínia i del 88,65% en la base diagonal, ambdues superiors al llinar del 70,71% requerit per a la violació de la desigualtat CHSH. El paràmetre S de CHSH s'ha avaluat experimentalment, obtenint $S = 2,41 \pm 0,008$, fet que suposa una violació del límit clàssic de 51,9 desviacions estàndard. Aquests resultats certifiquen que la font efectivament genera parells de fotons entrellaçats en polarització. Gràcies a la seva robustesa i estabilitat passiva intrínseques, la font es presenta com una opció prometedora per a futurs projectes de distribució quàntica de claus basats en satèl·lits.

*Dear Leibniz,
Natura facit saltus.*

Acknowledgements

I would like to express my most sincere gratitude to the many people whose guidance and support made this thesis possible.

First of all, I would like to deeply thank my supervisors, Prof. Dr. Valerio Pruneri and Dr. Robin Camphausen. To Valerio, for allowing me the opportunity to work in his group, explore a new research topic, and experience academia first-hand. To Robin, for the time and effort he devoted to supporting me throughout these past months, thank you so much for mentoring me even in this new and exciting new stage; I sincerely wish you many beautiful memories with your new family.

To Dr. Javier Argüello-Luengo, who readily accepted being my long-distance co-supervisor, I would like to express my thanks for fueling my passion for quantum technologies during his classes, along with Dr. Iacopo Torre. I am especially grateful for his constant encouragement to engage in activities beyond the classroom: sending us international opportunities and internships, and keeping us informed of conferences and outreach events, always going out of his way to encourage us to take part. This effort allowed me to grasp just how far quantum technology stretches, and helped me realize it is something I wholeheartedly want to be part of.

I would also like to thank my unofficial supervisors, Ilektra and Stefano, for taking the time to guide me these past months. I am especially grateful to Stefano for taking me under his wing: for his expert guidance, the countless hours he spent with me in the lab, and his insightful feedback and suggestions throughout this work.

A big acknowledgement to all the members of the Optoelectronics group at ICFO, who all contribute to creating a really special supportive and stimulating atmosphere, which it was a pleasure to work in. Their expertise and willingness to share their knowledge have been truly inspiring, and I also deeply appreciate all the trust and independence I was given, which allowed me to grow both personally and professionally.

The biggest thank you goes to my family, for their continued loving help and support, making anything possible. To my Mom and Dad, who feel as thrilled for my journey as I do, if not more, I would like to immortalize this recurring thought on paper: *moltes gràcies pel vostre suport incondicional.*

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1 Introduction

1.1 Motivation

The next technological breakthroughs in the upcoming years will likely draw on the principles of quantum physics [1]. Fields such as medicine [2, 3], computation [4], materials [5], or sensing [6] are already experiencing major transformations. In particular, communication is actively being redefined, revolutionizing how information is transmitted and secured.

Much of our current society’s infrastructure relies on secure data transmission, ranging from private communication to confidential data protection. The security of most deployed protocols hangs on the presumed computational hardness of certain mathematical problems, an assumption that is challenged by the advent of quantum computation [7]. The formulation of the Shor algorithm, a quantum algorithm capable of efficiently breaking widely used encryption schemes such as RSA or ECC [8], combined with recent experimental advances towards scalable fault-tolerant quantum computation [9], pose a genuine threat to the classical security paradigm. Fortunately, the remedy for this eroding framework also stems from quantum mechanics itself, providing solutions that rest on unconditional physical guarantees, rather than on presumed computational difficulty.

Currently, a well-established enabler for secure quantum-safe communications is quantum key distribution (QKD), which allows two parties to produce a shared secret key. Depending on which property is exploited, QKD has two main approaches [10]. *Prepare-and-measure* protocols (such as the BB84) are based on the no-cloning theorem and the measurement-disturbance principle. They offer higher communication rates and require simpler hardware requirements, typically relying on weak coherent pulses [11]. *Entanglement-based* protocols instead leverage non-local correlations to provide enhanced security and, since they do not require trust in the state-preparation device, they are resilient to attacks targeting source imperfection [12]. Moreover, the gold-standard for secure key exchange is device-independent QKD, which can only be achieved through the use of entanglement [13]. This second approach, however, is constrained by, on the one hand, the increased complexity of high-fidelity pair generation and synchronization over long-distance channel [14]; and secondly, the sensitivity to channel loss and noise, which induces loss and decoherence, impacting key rates [15].

Entanglement generation is thus a critical building block not only for device-independent QKD, but to the broader landscape of quantum information [16, 17]. This wide relevance motivates the focus of this work on entanglement and, specifically, on the development of an efficient, high-performance *entangled photon source* (EPS).

Photons are the physical system of choice for quantum communication. Being weakly coupled to the environment, photons are affected by limited decoherence, enabling quantum states of light at room temperature [18]. Moreover, photon qubits can be transmitted at high speed, benefiting from the already mature fibre-based communication network [19], and opening the possibility for future global quantum communication applications [20].

There are various types of entangled photon sources, depending on the leveraged degree of freedom: spatial mode, frequency-bin, energy-time, polarization, and time-bin. Due to polarization’s robustness to environmental interference in transmission such as fibre or free-space, polarization entanglement is the common choice [21]. Additionally, the polarization-encoded state is a two-level photonic system, thus having a natural, well-established qubit coding, as will be presented in Section 2.2.2 [22].

1.2 Entangled photon sources

EPSs have progressed in accordance to the increasing demands imposed by quantum information applications. Presently, the most widely employed approaches are probabilistic photon-pair sources based on spontaneous wave-mixing processes in nonlinear optical materials [23], which result in highly efficient and versatile generation schemes for entanglement in various photonic degrees of freedom. Typical wave-mixing techniques exploited in photon-pair sources are based on accessing either the second-order ($\chi^{(2)}$) or third-order ($\chi^{(3)}$) optical nonlinearities, which lead to spontaneous parametric down conversion (SPDC) or four-wave mixing regimes (FWM), respectively.

SPDC represents to date the most commonly used "workhorse" EPS technology, since it readily allows accessing various freedom degrees. Hence, it has been used to achieve entanglement in polarization [24], frequency [25], time [26] and spatial mode [27], and also hyper-entangled states [28]. As a result, SPDC is a popular choice for quantum information research [29]. Moreover, the supporting technology for SPDC implementation can be modified with ease, allowing for the study of various applications. As a result, bulk-based SPDC systems are a mature technology, compared to the alternatives presented hereafter. There are, however, major drawbacks: bulk SPDC has poor scalability potential, it is sensitive to mechanical and thermal drifts, and has poor scaling to many-modes prospects [23]. Consequently, it motivates research in other directions. An alternative approach are fibre-based and waveguide sources, which present the immediate advantage of being telecom compatible [30, 31]. Said compatibility is a vital feature: for practical quantum communication sources ought to operate around telecom bands, as it allows to harness the existing optical-fibre infrastructure rather than requiring dedicated transmission channels [32]. This requirement steered the field towards periodically-poled lithium niobate (ppLN) waveguides, whose second-order nonlinearity allow SPDC to be performed directly at telecom wavelengths, rather than requiring frequency conversion after generation [32, 31]. Fibre-based architectures can obtain better stability compared to bulk sources, while being compatible with many existing telecom network devices [23].

On the other hand, spontaneous four-wave mixing (SFWM), stemming from $\chi^{(3)}$ nonlinearity, is a highly competitive alternative to SPDC and $\chi^{(2)}$ materials, principally owing to its ease of implementation in photonic integration technologies.

Both SPDC and SFWM schemes are often paired with a Sagnac interferometer to generate polarization entanglement [33, 24], offering high fidelity and brightness. Through the Sagnac loop, the pump is split into two counter-propagating paths with orthogonal polarizations, thus coherently superposing two indistinguishable nonlinear processes, which are then recombined when exiting the loop. Upon recombination, the two probability amplitudes interfere coherently, producing an entangled state. Due to its inherent symmetry, the Sagnac offers phase stability [34], removing the need for additional compensation optics. As handicaps, the interferometer has a larger physical footprint, and the separation of the pump between the two paths means the wave mixing processes do not use the full pump power. Nevertheless, polarizing Sagnac configuration is one of the most common sources [23]. Furthermore, due to the advantages this configuration offers, hybrid schemes leveraging fibre-based setups paired with a bulk Sagnac have demonstrated performance close to that of the best bulk-sources, all while being more compact (for instance, [35] developed a periodically-poled waveguide bidirectionally pumped inside a bulk Sagnac interferometer). Notably, a fully guided-wave source is also possible using the Sagnac loop [36], overcoming the bulk hindrance and allowing for further compactness.

1.3 State of the art of integrated polarization-entangled photon sources

The growing demand for compact, stable and scalable sources present photonic integration as the natural next step. *Photonic integrated circuits* (PICs) provide the attributes that bulk and fibre-based sources struggle to deliver simultaneously: a small footprint, mechanical robustness, and mass-manufacturability potential. Admittedly, historically PICs have not outperformed their bulk counterparts in achievable fidelity and efficiency of the source, mainly due to chip-coupling and waveguide-propagation losses. This trade-off gap is bound to gradually close as fabrication process mature, pushed by the worldwide investments in quantum technologies and quantum integrated photonics [23, 19, 37].

Multiple material platforms for the on-chip polarization-entangled photon generation have been put forward. For starters, lithium niobate offers strong $\chi^{(2)}$ nonlinearity, allowing for SPDC processes with excellent fidelities [38]. In particular, thin-film lithium niobate (TFLN) allows for architecturally simple designs, as [39] very recently showed: polarization entanglement can be generated within a single waveguide, without requiring interferometers or auxiliary polarization handling circuits that increase its complexity. This approach, however, relies on specialized fabrication steps, constraining its compatibility with complementary metal-oxide-semiconductor (CMOS) processes and thereby limiting its scalability and mass manufacturing potential.

Another material platform is AlGaAs, unusually combining $\chi^{(2)}$ and $\chi^{(3)}$ nonlinearities. It offers the prospect of monolithic integration with active components such as lasers, modulators, and detectors directly on-chip [40]. As an example, exploiting the third-order nonlinearity Steiner et al. [41] proposed an AlGaAs-on-insulator microring resonator generating pairs via SFWM, reaching a nearly 1000-fold brightness improvement over prior sources whilst maintaining high fidelities. Moreover, AlGaAs sources are compatible with direct electrical injection, facilitating their miniaturization [42]. Regrettably, such versatility comes at a cost. AlGaAs schemes suffer substantial polarization-dependent propagation loss, rendering in some cases unsuitable for integration with subsequent circuits [43]. Furthermore, despite AlGaAs can be integrated with silicon PICs, it is not natively CMOS-compatible, inhibiting the prospects for low-cost production [44].

Research focused on platforms such as TFLN and AlGaAs remains extremely active. Each offers a higher nonlinear efficiency than silicon, and, particularly in TFLN, achieve higher fidelities than on-chip in silicon up-to-date. That being said, the choice of the platform cannot rest on figures of merit alone. Envisioning a future quantum communication network, sources must be able to distribute entanglement on demand to multiple users, which in turn calls for broadband, compact sources, deployable at the scale and cost that only a mature, CMOS-compatible fabrication process can realistically offer [19, 45].

The silicon photonics platform, with its low costs and unique properties, is a paramount enabler of everyday applications stemming from fundamental research [46]. Its high refractive index contrast allows for highly compact waveguides, and the resulting optical density in small cores lowers the required pump power to trigger nonlinear effects. Most notably, silicon is inherently compatible with telecom communication techniques and CMOS fabrication [47], sharing the same manufacturing process that has driven the microelectronics industry for decades [19]. In the this context, it is logical to prioritize a mature, low-cost, wafer-scale platform that is seamlessly integrable with the existing telecommunications infrastructure [48].

For the reasons presented, the *silicon-on-insulator* (SOI) platform presents as the natural choice and, consequently, is the one employed in this work. As will be discussed in Section 2.3, due to the lack of second-order susceptibility in silicon, FWM is the most investigated and developed process on the SOI platform [19, 47, 21].

In view of $\chi^{(3)}$ nonlinearities yielding lower powers, microring resonators (MRRs) are a particularly attractive geometry for SFWM sources, yielding significantly higher brightness per unit bandwidth than an equivalent single-pass waveguide [19]. Regrettably, because of silicon waveguide’s birefringence, the transverse-electric (TE) and transverse-magnetic (TM) modes experience different effective indices and thus resonate at different frequencies. This entails that, without any additional devices, a single unidirectional MRR paired with SFWM results in frequency-bin entangled photon pairs, not the polarization entanglement this work requires [45].

To overcome this, several strategies have been proposed. The first on-chip polarization EPS on a SOI platform was put forward by Matsuda et al. [22], integrating a single-pass waveguide and a monolithic polarization rotator. As no cavity is involved, the non-degeneracy issue is avoided, but it forfeits the brightness advantage that motivated the use of a microring in the first place.

A second approach, brought by Zhang et al [43] employs two separate, independent MRRs, designed so that one resonates on TE and the other on TM. Their resonances are then spectrally aligned by design and by active thermal tuning which, thus preserving brightness. However, since the two rings are physically distinct resonators, they slightly differ in their coupling and propagation loss characteristics. This mismatch, along with any drift between their relative phase, degrades the fidelity of the resulting entangled state. Zhang et al. propose to mitigate the phase fluctuation by adopting an active phase-stabilization scheme, which would further increase the design’s complexity.

A third strategy returns to a single, bidirectionally-pumped MRR, whose counter-propagating SFWM processes are combined via a Sagnac loop to produce polarization entanglement (which is described in further detail in Section 2.3.4). Until recently, the Sagnac configuration had been mostly implemented off-chip in a hybrid architecture [49], benefiting from the phase stability this configuration provides, albeit sacrificing compactness and scalability. To address this remaining off-chip dependency, Miloshevsky et al. [45] and Hua et al. [21] integrate a polarization splitter-rotator (PSR) on-chip to combine two bidirectional SFWM amplitudes.

1.4 Motivation and scope of this work

This work presents a polarization-entangled photon-pair source that is monolithically integrated on a SOI platform. A PSR is paired with a Sagnac loop to generate entanglement through SFWM, while the silicon waveguide preserves the polarization state on-chip, removing the need for off-chip polarization control.

As earlier outlined, recent demonstrations of integrated polarization-entangled sources, such as the PSR-based Sagnac loop of Hua et al. [21], have relied on MRRs to enhance SFWM efficiency through field confinement. This choice, however, introduces a practical limitation: MRRs are thermally sensitive, and must be actively stabilized against temperature fluctuations to remain on resonance with the pump. This requirement nullifies one of the principal advantages of SFWM over SPDC: its compatibility with standard CMOS fabrication without the need for the precise temperature control that SPDC sources typically require. To date, no monolithically integrated, polarization-entangled SFWM source has been demonstrated that dispenses with resonant enhancement entirely and is passively stable by design.

The presented work addresses that gap. By using a bare, non-resonant Sagnac loop rather than a microring-based one, such that no active thermal control is required to stabilize the source, albeit losing the advantages MRRs provide. Moreover, the source is optimized for robustness in applications in harsh environments, such as satellite-based

links, where the device must be passively stable.

1.5 Thesis objectives

This thesis is dedicated to characterizing a monolithically integrated polarization-entangled photon pair source on a SOI platform. To this end, the main objectives of this work are:

- Construction of the PIC probe-station experimental setup: designing and assembling a dedicated characterization bench capable of executing both classical and quantum optical measurements, aiming for stability and coupling accuracy.
- Classical characterization of the chip: this includes measuring waveguide propagation and coupling losses, extinction ratios of the integrated polarization management components, and stimulated four-wave mixing (sFWM) efficiency.
- Quantum light generation and spectral properties to generate and characterize quantum states of light produced via on-chip spontaneous four-wave mixing (SFWM), evaluating the brightness and coincidence-to-accidental ratio (CAR).
- Certification of polarization entanglement to assess the polarization correlations by measuring two-photon polarization interference fringes and evaluating their visibility. The entanglement certification is further supported by experimentally showing the violation of the CHSH inequality.

This work is organized as follows. Chapter 2 introduces the theoretical background of quantum information, photonic quantum states, and nonlinear silicon photonics, focusing on spontaneous four-wave mixing and Sagnac-loop geometries. Chapter 3 details the experimental methodology, describing the silicon photonic integrated circuit fabrication, the probe-station setups, and the protocols used for classical and quantum characterization. Chapter 4 presents the experimental results, including loss measurements, stimulated four-wave mixing efficiencies, coincidence-to-accidental ratios, and entanglement certification via two-photon interference and CHSH inequality violations. Finally, Chapter 5 offers conclusions and future perspectives, while Chapter 6 provides an evaluation of the project's sustainability and ethical implications.

2 Theoretical framework

2.1 Basics of quantum information

A pivotal moment in modern physics was the realization that classical mechanics, for all its success, fails to describe how the world behaves at the smallest scales. Classical theory assumes a strict dichotomy between waves and particles, alongside a fully deterministic description of physical phenomena. However, neither holds true under experimentation. [50].

Fortunately, quantum mechanics reconciles these apparently conflicting notions by introducing the concept of *wave function*. In this framework, measurement outcomes are inherently probabilistic, and randomness is taken as a fundamental postulate, rather than being a consequence of incomplete knowledge [51, 52]. This gives rise to crucial features of quantum mechanics, namely interference, and entanglement.

This chapter begins with a brief overview of the quantum states of light, with particular attention on entanglement and projective measurements; we then devote our attention to basic non-linear optics concepts that are paramount for the experiments, namely the four-wave mixing phenomenon, and a brief survey of the different geometries through which it can be implemented.

2.1.1 Quantum state vector

Let $\mathcal{H}$ be a Hilbert space, which will be assumed to be complex and separable. In quantum mechanics, a pure state is represented by a vector $|\Psi(\mathbf{r}, t)\rangle \in \mathcal{H}$, hereafter simply denoted by $|\Psi\rangle$, as we shall work with finite-dimensional degrees of freedom (e.g. polarization) rather than the spatial wavefunction. Said state must be normalized, and defined up to a global phase. Similarly, observables correspond to operators acting on $\mathcal{H}$. [53]

Given a set of vectors $\{|\psi_i\rangle\}$ acting as a complete orthonormal basis of $\mathcal{H}$, the state can be written as

$$|\Psi\rangle = \sum_{i=1}^{\infty} c_i |\psi_i\rangle, \quad (2.1)$$

where $c_i = \langle\psi_i|\Psi\rangle \in \mathbb{C}$, and $\sum_i |c_i|^2 = \langle\Psi|\Psi\rangle = 1$.

However, not all physical systems can be described by a single state vector: those that cannot are known as mixed states. To provide a more general representation of the states, the *density matrix* is used [54],

$$\hat{\rho} = \sum_i p_i |\phi_i\rangle \langle\phi_i|, \quad (2.2)$$

where $|\phi_i\rangle \in \mathcal{H}$ are normalized states (not necessarily orthogonal) with weights $p_i \geq 0$ satisfying $\sum_i p_i = 1 \iff \text{Tr}\{\hat{\rho}\} = 1$. Through the density matrix formalism one can define the purity of a state as

$$\mathcal{P} = \text{Tr}\{\hat{\rho}^2\}, \quad (2.3)$$

satisfying $0 < \mathcal{P} \leq 1$, where $\mathcal{P} = 1 \iff \hat{\rho} = \hat{\rho}^2$ corresponds to a pure state, and otherwise $\mathcal{P} < 1$ and it is mixed.

2.1.2 Entanglement and separability

A quantum composite system, or *multipartite system*, can be decomposed into two or more quantum subsystems. Let $\mathcal{H}$ be the Hilbert space associated with a multipartite system, it

is given by the tensor product of the spaces of its parts: $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3 \otimes \dots$ [55]. Note that, hitherto, no restriction has been placed on the dimension of $\mathcal{H}$. For the purpose of this work, we shall henceforth focus on the 2-dimensional, as is where our measurements reside.

Thus, let us consider the bipartite system $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ in which each subsystem is a pure state $|\psi_i\rangle$ ($i = 1, 2$). The composite system is trivially:

$$|\Psi_s\rangle = |\psi_1\rangle \otimes |\psi_2\rangle. \quad (2.4)$$

In this case, the measurement on one subsystem (local measurement) does not affect the outcome on the other one: they are uncorrelated. This is denoted as a separable state.

One can define a general pure state given by the superposition of the states of the form (2.4),

$$|\Psi_e\rangle = \frac{1}{\sqrt{2}}(|\psi_1\rangle \otimes |\psi_2\rangle + |\phi_1\rangle \otimes |\phi_2\rangle), \quad (2.5)$$

where $|\psi_i\rangle \neq |\phi_i\rangle$. In this case, $\nexists |\psi_i\rangle \in \mathcal{H}_1, |\psi_j\rangle \in \mathcal{H}_2$ such that $|\Psi_e\rangle = |\psi_i\rangle \otimes |\psi_j\rangle$. That is, local measurement on one subsystem will condition the outcome of the other subsystem, hence it is not a separable state.

In general, states that can be written as a product of pure states, as in Eq. (2.4) are called *separable states*. Contrarily, if there are no local states the product of which define the system $|\Psi\rangle \in \mathcal{H}$, then $|\Psi\rangle$ is an *entangled state* [55, 54, 19]. We also introduce the notion of a bipartite pure state being maximally entangled, where the conforming subsystems are maximally entangled on their own. This concept will be useful later on.

2.1.3 Qubits and Bell states

In classical information theory, information is encoded into the bit, with possible values 0 or 1. Its quantum counterpart is the *qubit*, a bidimensional quantum state that can not only represent pure 0 and 1 states, but also their superposition: $|\psi\rangle = c_0|\mathbf{0}\rangle + c_1|\mathbf{1}\rangle$, satisfying $|c_0|^2 + |c_1|^2 = 1$ and $\langle\mathbf{0}|\mathbf{1}\rangle = 0$ [50]. Qubits are used to store quantum information, which is the basis for quantum computation and quantum communication.

One can find a specific class of entangled quantum state, which are paramount in quantum information: the *Bell states*. For simplicity, let us introduce the notation $|\phi\psi\rangle \equiv |\phi\rangle \otimes |\psi\rangle$, for two arbitrary states $|\phi\rangle, |\psi\rangle$. Then, the Bell states are defined as follows:

$$|\Phi^\pm\rangle_{AB} = \frac{1}{\sqrt{2}}(|\mathbf{00}\rangle_{AB} \pm |\mathbf{11}\rangle_{AB}) \quad (2.6)$$

$$|\Psi^\pm\rangle_{AB} = \frac{1}{\sqrt{2}}(|\mathbf{01}\rangle_{AB} \pm |\mathbf{10}\rangle_{AB}), \quad (2.7)$$

where the subscripts A, B indicate which subsystem the basis states belong to (that is, $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$). States (2.6) and (2.7) are maximally entangled states, and form a maximally entangled basis, the Bell basis [53].

2.1.4 Projective measurements and change of basis

A qubit admits infinitely many orthonormal bases in which it can be expanded, and the physical predictions of quantum mechanics depend on which basis a measurement is performed in. Besides the computational basis $\{|\mathbf{0}\rangle, |\mathbf{1}\rangle\}$ previously introduced, another com-

monly used is the Hadamard basis,

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle), \quad (2.8)$$

$$|-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle), \quad (2.9)$$

which is itself orthonormal and complete, and is related to the computational basis by a 45° rotation [53].

Measuring a qubit in a given basis is described by a *projective measurement*: given an orthonormal basis $\{|u_i\rangle\}$, the probability of obtaining outcome i on a state $|\psi\rangle$ is $\mathcal{P}(i) = |\langle u_i|\psi\rangle|^2$, and the post-measurement state collapses onto $|u_i\rangle$. Crucially, a state that is *definite* in one basis is generally a *superposition* in another. For instance, the computational basis state $|0\rangle$, when expressed in the Hadamard basis, reads

$$|0\rangle = \frac{1}{\sqrt{2}} (|+\rangle + |-\rangle), \quad (2.10)$$

so measuring $|0\rangle$ in the $\{|+\rangle, |-\rangle\}$ basis yields either outcome with equal probability $1/2$, despite $|0\rangle$ being a pure state. This equal split is a coherent superposition, and is reversible: measuring back in the computational basis returns $|0\rangle$.

This basis-dependence has significant physical repercussions once a *relative phase* is introduced. Let us consider the general single-qubit state:

$$|\psi(\phi)\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\phi} |1\rangle), \quad (2.11)$$

which for any ϕ yields outcome probabilities $\mathcal{P}(0) = \mathcal{P}(1) = 1/2$ in the computational basis. However, if instead it is expressed in the Hadamard basis,

$$|\psi(\phi)\rangle = \frac{1}{2} [(1 + e^{i\phi}) |+\rangle + (1 - e^{i\phi}) |-\rangle], \quad (2.12)$$

so that

$$\mathcal{P}(+) = \frac{1 + \cos \phi}{2}, \quad \mathcal{P}(-) = \frac{1 - \cos \phi}{2}. \quad (2.13)$$

The relative phase ϕ manifests as an oscillatory *interference* pattern in the Hadamard basis, despite being entirely invisible to a computational-basis. Indeed, the two amplitudes in Eq. (2.11) interfere constructively or destructively depending on ϕ , much like two coherent waves would. Through this mechanism, phase information is converted into probability information by changing the measurement basis, and it underlies the correlation fringes measured for the polarization Bell states relevant to this work.

2.1.5 Bell's theorem and the CHSH inequality

The introduction of the concept of entangled state led to relevant discussions on the nature of this phenomenon. Famously, in 1935 the physicists A. Einstein, B. Podolsky and N. Rosen published a paper, the so-called EPR paradox, in which they challenged the Copenhagen interpretation of quantum theory [56]. They proposed that quantum mechanics is an incomplete theory, and that the supposedly probabilistic character of the measurement outcomes are in fact determined by hidden variables, not accounted for by the quantum formalism [57]. Specifically, they must be *local* hidden variables (LHV): the outcome of a subsystem's measurement must be independent from the measurement setting performed

on a distant, non-interacting subsystem. In contrast with the LHV approach, in the Copenhagen interpretation the entangled state wave function is intrinsically non-local prior to measurement [58].

In 1964, John Bell designed a variation on the EPR experiment and proved that the LHV and quantum mechanical theories do not in fact provide indistinguishable outcomes, but can actually predict different results [59], showing that microscopic systems are non-local by nature. Bell's key result was the derivation of the so called *Bell's inequality*, which any LHV theory must satisfy, whereas quantum mechanics predicts its violation in certain scenarios. This incompatibility in both theories is showcased in Bell's theorem, valid on all LHV models [50].

Despite there being several versions of Bell's theorem, perhaps the most widely experimentally used in experiments is the one derived by Clauser, Horne, Shimony, and Holt (CHSH) in 1969 [60]. They introduced an experimentally determinable parameter S defined by:

$$S = E(\theta_a, \theta_b) - E(\theta_a, \theta'_b) + E(\theta'_a, \theta_b) + E(\theta'_a, \theta'_b), \quad (2.14)$$

where θ_a, θ'_a and θ_b, θ'_b denote the measurement angles (detector settings) on sides A and B , respectively. This correlation function, $E(\theta_a, \theta_b)$ of the pairs is given by

$$E(\theta_a, \theta_b) = \mathcal{P}_{00}(\theta_a, \theta_b) + \mathcal{P}_{11}(\theta_a, \theta_b) - \mathcal{P}_{01}(\theta_a, \theta_b) - \mathcal{P}_{10}(\theta_a, \theta_b), \quad (2.15)$$

where $\mathcal{P}_{ij}(\theta_a, \theta_b)$ denotes the (normalized) coincidence probability of obtaining outcomes $i, j \in \{0, 1\}$ on sides A, B respectively. Bell proved that the following Bell inequality:

$$-2 \leq S \leq 2, \quad (2.16)$$

holds for all possible LHV theories, and nonetheless is routinely violated by quantum experiments. Furthermore, the CHSH inequality is also constrained with an upper bound, the Tsirelson's bound: $|S| \leq 2\sqrt{2}$ [61].

A measured value of $|S| > 2$ constitutes an experimental violation of the CHSH inequality, thereby certifying that the photonic source is indeed entangled.

2.2 Photonic quantum states of light

Section 2.1 introduced some basic concepts to represent the quantum state of a system, without making reference to any particular physical implementation. Here, we show how different quantum states can be realized by photons.

2.2.1 Photon number state

Quantum theory of light is based on the quantum harmonic oscillator (QHO), where the energy levels of a given electromagnetic field are quantized. The ground state of the QHO represents the lowest-energy configuration, denoted as $|0\rangle$. The operators that increase or decrease the eigenstate are the creation and annihilation operators, denoted by convention as $\hat{a}^\dagger$ and $\hat{a}$, respectively. By applying the creation operator to the ground state a single electromagnetic quanta is generated: a photon [54].

Such ladder-like treatment naturally leads to the number state representation, in which each eigenstate is labelled as $|n\rangle$, the number state or *Fock state*. This way, given the Hamiltonian of a harmonic oscillator $\hat{H}$, the energies will be given by $\hat{H}|n\rangle = E_n|n\rangle = (n + \frac{1}{2})\hbar\omega$, and the application of the creation operator $\hat{a}^\dagger$ yields a new eigenstate of energy

$E_n + \hbar\omega$ [50]. Consequently, we can define the following operations:

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle \quad (2.17)$$

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle, \quad (2.18)$$

for the creation operator and annihilation operators, respectively. We can whence express the Fock state as follows:

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle. \quad (2.19)$$

Let us define the photon number operator, $\hat{N} = \hat{a}^\dagger \hat{a}$, which satisfies $\hat{N} |n\rangle = n |n\rangle$, and we can thus easily obtain the number of photons of a state: $\langle n | \hat{N} | n \rangle = n$.

Interestingly, the photon number representation can be used to define a coherent state $|\alpha\rangle$:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (2.20)$$

The expectation value yields the mean photon number of a coherent state: $\langle \alpha | \hat{N} | \alpha \rangle = |\alpha|^2 = \bar{n}$. This result agrees with the classical notion of the intensity of a coherent electromagnetic field. Finally, it is useful to compute the probability that there are n photons in a coherent state,

$$\mathcal{P}(n) = |\langle n | \alpha \rangle|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!} = \frac{\bar{n}^n}{n!} e^{-\bar{n}}. \quad (2.21)$$

Note that Eq. (2.21) is a Poisson distribution with variance $\bar{n}$, therefore concluding that coherent states have an uncertainty in photon number with $\sigma = \sqrt{\bar{n}}$. Poissonian photon statistics cause *shot noise* in photodetection [54, 50, 51].

2.2.2 Polarization degree of freedom

Qubits are realized via physical systems that meet the requirements presented in 2.1.3. In rough terms, an individual quantum system with two clearly distinguishable and measurable states. *Polarization-encoded qubits* are a commonly adopted approach, due to their robustness in transmission and the readily available tools for state manipulation [62, 63]. In particular, we shall focus on linear polarization.

Let us begin by defining the $|H\rangle$ and $|V\rangle$ as the following single photon Fock states:

$$|H\rangle = \hat{a}_H^\dagger |0\rangle, \quad (2.22)$$

$$|V\rangle = \hat{a}_V^\dagger |0\rangle, \quad (2.23)$$

where the H (V) subscript on the $\hat{a}^\dagger$ operators indicates that the photon is created in the horizontal (vertical) polarization mode. As they are orthogonal to each other, $|H\rangle$ and $|V\rangle$ make for natural qubit basis vectors, that is: $|0\rangle \equiv |H\rangle$, $|1\rangle \equiv |V\rangle$. Furthermore, it is useful to define the diagonal and anti-diagonal linear polarization states, respectively:

$$|D\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle), \quad (2.24)$$

$$|A\rangle = \frac{1}{\sqrt{2}}(|H\rangle - |V\rangle). \quad (2.25)$$

Whichever direction is taken as "horizontal" and "vertical" is arbitrary, however typical practice in laboratory experiments is to designate the direction parallel to the table surface as "horizontal" [54].

Recalling the definition of Bell states, in Eqs. (2.6) and (2.7), it is clear that they can be realized with two entangled polarized photons. For instance,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|HH\rangle_{AB} + |VV\rangle_{AB}), \quad (2.26)$$

where photons A and B are entangled in their polarization degrees of freedom. Although photons are bosons and therefore a two-photon state is symmetrical upon exchange (as they do not obey Pauli's exclusion principle), it is useful to assign labels A, B when said pair is distinguishable in some other degree of freedom besides polarization –let it be spatial mode, time-bin, frequency, or orbital angular momentum. This way, it is possible to track them separately [54, 50].

2.2.3 Polarization basis projection and Bell-state correlations

The polarization qubit just outlined in the preceding section follows exactly the $\{|0\rangle, |1\rangle\}$ and $\{|+\rangle, |-\rangle\}$ structure of Section 2.1.4, with $|H\rangle, |V\rangle$ playing the role of the computational basis and $|D\rangle, |A\rangle$ the Hadamard basis. In this section we shall dive into this relation, as will be of utmost importance for the findings of this work.

Consider first the separable reference state $|HH\rangle = |H\rangle_A |H\rangle_B$. Expressing each photon's polarization in the diagonal basis as

$$|H\rangle = \frac{1}{\sqrt{2}}(|D\rangle + |A\rangle), \quad (2.27)$$

$$|V\rangle = \frac{1}{\sqrt{2}}(|D\rangle - |A\rangle), \quad (2.28)$$

results in:

$$|HH\rangle = \frac{1}{2}(|DD\rangle + |DA\rangle + |AD\rangle + |AA\rangle). \quad (2.29)$$

As such, a D/A -basis measurement of this state is expected to yield all four coincidence outcomes with equal probability, $\mathcal{P}(DD) = \mathcal{P}(DA) = \mathcal{P}(AD) = \mathcal{P}(AA) = 1/4$. Recall that this equal split does not indicate an incoherent mixture ($|HH\rangle$ remains a pure product state), it simply reflects that H has equal projection onto D and A . Consequently, the diagonal basis carries no information distinguishing $|HH\rangle$ from any other separable state of this form, and serves as a useful control case for the entangled example presented hereafter.

Now, let us consider the target Bell state,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|HH\rangle + |VV\rangle). \quad (2.30)$$

Applying Eq. (2.28) to both terms, the cross terms cancel and $|\Phi^+\rangle$ reduces to

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|DD\rangle + |AA\rangle), \quad (2.31)$$

so that, unlike the separable reference state, Φ^+ predicts $\mathcal{P}(DD) = \mathcal{P}(AA) = 1/2$ while $\mathcal{P}(DA) = \mathcal{P}(AD) = 0$. This means that the polarization correlation observed in the H/V basis is also present in when rotated 45° .

Notice how this is precisely the phase-interference mechanism presented in Section 2.1.4. Indeed, let us consider the more general state:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|HH\rangle_{AB} + e^{i\phi}|VV\rangle_{AB}). \quad (2.32)$$

The diagonal basis correlation depend on the relative phase ϕ , with $\phi = 0$ resulting in Φ^+ for which the co-polarized correlations are fully preserved under basis rotation. As a result, observing stable correlations in both rectilinear and diagonal basis is necessary to ensure that a source is producing the desired maximally entangled state (with $\phi = 0$), not merely two independently, incoherently generated $|HH\rangle$ and $|VV\rangle$ contributions that happen to mimic Φ^+ -like statistics in a single basis.

More generally, projecting $|\Phi^+\rangle$ onto arbitrary analyzer angles θ_A, θ_B yields

$$\mathcal{P}_{\text{same}}(\theta_A, \theta_B) = \frac{1}{2} \cos^2(\theta_A - \theta_B), \quad (2.33)$$

$$\mathcal{P}_{\text{cross}}(\theta_A, \theta_B) = \frac{1}{2} \sin^2(\theta_A - \theta_B), \quad (2.34)$$

for the probability of obtaining equal or orthogonal projections, respectively, where $\mathcal{P}_{\text{same}} = \mathcal{P}(\theta_A, \theta_B) + \mathcal{P}(\theta_{A\perp}, \theta_{B\perp})$ and similarly for $\mathcal{P}_{\text{cross}}$. Therefore, the coincidence rate on any given channel pair oscillates sinusoidally as a function of the *relative* analyser angle, $\theta_A - \theta_B$ —not as a function of θ_A, θ_B individually. This signature characteristic allows to discern Φ^+ from any separable state (for which the two analyser settings do contribute independently).

Moreover, Eqs. (2.33) and (2.34) are related to the CHSH correlation function of Section 2.1.5, identifying the $\{0, 1\}$ outcomes with the $\theta/\theta_\perp$ projections as follows:

$$E(\theta_A, \theta_B) = \mathcal{P}_{\text{same}} - \mathcal{P}_{\text{cross}} = \cos(2(\theta_A - \theta_B)), \quad (2.35)$$

a polarization form of the correlation function that defines the CHSH parameter S .

The presented mechanism conforms the basis of the experimental procedure this work is based on.

2.3 Nonlinear silicon photonics

Silicon has emerged as the leading platform for integrated photonics, thanks to its compatibility with the mature complementary metal-oxide-semiconductor (CMOS) technology used in modern microelectronics, which paves the way towards potential mass manufacturability and low-cost integration. In addition, the high refractive index contrast between silicon and its surrounding materials enables for strong optical confinement, allowing for the compactness of the optical devices [46].

It is precisely these properties that make silicon particularly attractive for nonlinear optical applications. It exhibits a high third-order nonlinear susceptibility and strong mode confinement, which lead to high optical intensity within waveguides, thus reducing the power required to trigger nonlinear effects.

2.3.1 Third-order susceptibility

Let $\tilde{E}(t)$ be an applied electric field. In linear optics, the induced polarization $\tilde{P}(t)$ depends on the electric field following $\tilde{P}(t) = \varepsilon_0 \chi^{(1)} \tilde{E}(t)$, where ε_0 is the free space permittivity and $\chi^{(1)}$ the linear susceptibility. In nonlinear optics, the previous equation is generalized [64],

$$\tilde{P}(t) = \varepsilon_0 \left[\chi^{(1)} \tilde{E}(t) + \chi^{(2)} \tilde{E}^2(t) + \chi^{(3)} \tilde{E}^3(t) + \dots \right]. \quad (2.36)$$

The quantities $\chi^{(2)}$ and $\chi^{(3)}$ are known as the second- and third-order nonlinear optical susceptibilities, respectively, and they are third- and forth-order tensors [65].

Although second-order nonlinear processes generally operate under weaker fields than third-order ones, their implementation in integrated photonics remains challenging. In

particular, $\chi^{(2)}$ materials are not generally CMOS-compatible, rendering them costly to integrate. By contrast, silicon possesses a centrosymmetric crystalline structure, and therefore inversion symmetry, which prevents the existence of even-order nonlinearities – i.e. $\chi^{(2)} = 0$. Consequently, silicon is governed by the third-order susceptibility. Moreover, being $\chi^{(3)}$ a fourth-order tensor, the nonlinear response will be dependent on the polarization state of the interacting fields [65, 19].

The third-order optical nonlinearity gives rise to a rich variety of processes. For instance, self- and cross-phase modulation, in which the refractive index becomes intensity-dependent (also known as the optical Kerr effect [66]); two-photon absorption (TPA), in which two photons are absorbed simultaneously across the material bandgap; and four-wave mixing (FWM), where four waves interact through an energy exchange process [67].

At telecom wavelengths, particularly at 1550 nm, Silicon exhibits non-negligible TPA, which will be revisited as a limiting noise factor both in terms of maximum usable power and noise [19]. Finally, amongst the presented processes, we shall turn our attention to four-wave mixing in the subsequent section, as is the one exploited in this work to generate the photon pairs.

2.3.2 Four-wave mixing in Silicon waveguides

Four-wave mixing (FWM) is the most exploited nonlinear process on the SOI platform [68, 69]. It is a *parametric* process, meaning that the quantum state of the medium is left unchanged, and the medium population can be removed from the ground state only briefly, when it resides in a virtual level [65]. In contrast, an example of a non-parametric process would be Raman scattering, which results from the interaction between propagating photons and vibrational modes of the material structure [19]. Back to FWM, Figure 1 shows its most simple configuration, two input photons, hereafter referred to as *pump* photons, at the same frequency ω_p are annihilated to simultaneously generate a *signal* and *idler* photons, of frequencies ω_s and ω_i , respectively, with $\omega_i > \omega_s$.



Figure 1: Schematic representation of a FWM process in a waveguide.

In addition to the pump we can also provide a seed signal, injected at the signal or idler frequency. In this case, the process is said to be *stimulated* FWM (sFWM). By adding the seed we stimulate the generation process, hence boosting the generation rate of signal-idler pairs. By contrast, when no stimulus is provided we are dealing with the *spontaneous* FWM (SFWM) [19, 65].

Both sFWM and SFWM are subject to the same two constraints. Firstly, by the energy conservation condition, $2\omega_p = \omega_s + \omega_i$, which forces the signal and idler frequencies to be symmetric about the pump frequency. The second constraint is the momentum conservation, which arises from the phase-sensitive nature of the parametric interaction: the generated fields in the waveguide interact constructively provided that they are generated in phase, the so-called *phase-matching condition*,

$$\Delta k = 2k_p - k_s - k_i, \quad (2.37)$$

with $k = \frac{\omega}{c}n_{eff}$ the wave vector, n_{eff} the effective index, and Δk the phase-mismatch, which will yield zero when the phase-matching condition is met, maximizing the FWM efficiency.

It must be noted that in nonlinear processes the generated fields do not necessarily build up coherently along the propagation direction. Their relative phase evolution is governed by the geometry-dependent dispersion of n_{eff} , and can therefore be engineered through the waveguide cross-section to promote constructive interference [48]. Let L be the length of the waveguide. Provided that phase-matching is satisfied, the efficiency η of FWM can be written as:

$$\eta \propto \text{sinc}^2 \left(\Delta k \frac{L}{2\pi} \right). \quad (2.38)$$

Stimulated FWM can be fully described within a classical electromagnetic framework. This is no longer the case for spontaneous FWM: with no input field to seed the process, classical theory would predict a vanishing conversion rate, at odds with what is experimentally observed. Thus, we require a quantum description. In this case, the seed is provided by the vacuum quantum fluctuations [70], and the process is characterized by the fact that the generated signal and idler photons are always emitted simultaneously in pairs and correlated in time. The degree of said correlation is set by the joint bandwidth of the pump field and of Eq. (2.38) [48].

It is precisely this pair correlation, arising on-chip from the $\chi^{(3)}$ nonlinear interaction, that constitutes the photon-pair source exploited in this work. Together with the Sagnac-loop geometry described in Section 2.3.4, it provides the means to generate polarization-entangled photon pairs.

2.3.3 Limits to spontaneous FWM efficiency

The same third-order nonlinearity responsible for SFWM is also responsible for several side processes that degrade the quality of the generated photon pairs, by inducing losses or by limiting the source's single photon quality [19].

Silicon is strongly affected by TPA at 1550 nm, generating free carriers within the waveguide. These free carriers give rise to extra absorption and dispersion phenomena, named respectively free-carrier absorption (FCA) and free-carrier dispersion (FCD). Both act on the pump and on any generated signal, thus limiting the maximum pump power usable on chip and inducing extra losses on the generated photon pairs [46].

Another nonlinear noise source is spontaneous Raman scattering from the waveguide itself, in which a pump photon is inelastically scattered by a phonon of the silicon lattice. This gives rise to a well-defined peak distinguishable from the pump's, and the scattered photons are emitted independently, hence are uncorrelated. Nonetheless, because the Raman shift is fixed, this noise contribution can be minimized by design [19].

Finally, at sufficiently high pump power, multipair generation becomes non-negligible: the probability of generating more than one signal-idler pair within the same generation event becomes high. These multipair events are statistically independent from one another, and therefore introduce uncorrelated coincident events and degrade the desired single-pair character of the source [71]. This phenomenon increases with pump power, so we are forced to work with lower power regimes, trading off generation rate against pair quality [72], [19].

The presented processes constitute the main physical mechanisms that contribute to limiting the performance of the source.

2.3.4 Spontaneous FWM photon-pair generation geometries

The simplest SFWM implementation to generate photon-pairs consists of a single-pass waveguide with an unidirectional pump, generating signal-idler pairs at the output. This straightforward geometry has a caveat: it does not produce a polarization-entangled state.

We require an interferometric configuration to coherently combine two indistinguishable generation processes. In other words, the circuit must allow the introduction of a second generation path, indistinguishable from the first one.

One approach is to rely on *post-selection*. Here, the source produces pairs through two processes from a single-pass source, which may be distinguishable (for instance, by which spatial path the photon travels, or difference in arrival time). By conditioning on a measurement outcome, say, coincidence detection, only the detection events for which the which-path information can't be recovered are kept. This remaining ensemble corresponds to a coherent superposition of the two processes, yielding an entangled signal-idler state [23]. This approach forces to discard a significant fraction of the generated pairs, rendering it unsuitable for applications where high collection efficiency is needed.

The second strategy generates both polarization components simultaneously and independently, removing the need for post-selection. This can be achieved via microring resonators that generate orthogonally polarized resonances [43], though the details will be spared. The geometry adopted in this work, and the most widely used in integrated SFWM sources, is the *Sagnac loop*. A polarization-selective element splits the pump into clockwise (CW) and counter-clockwise (CCW) beams, which enter the same waveguide before recombining at the same element [32]. As they share the same physical path, both directions share resonant frequencies, and any phase fluctuations or thermal drifts will affect them equally: we have two independent SFWM processes happening simultaneously.

Polarization entanglement is achieved by making the CW- and CCW-generated pairs distinguishable in polarization, and so are made to emerge from the loop in orthogonal polarizations (Section 3.2.2 will cover how such orthogonality is induced). Since processes are indistinguishable in any other aspect, their coherent recombinations yields the state

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|HH\rangle_{AB} + e^{i\phi}|VV\rangle_{AB}), \quad (2.39)$$

equivalent, up to the relative phase ϕ due to the loop birefringence, to the Bell state $|\phi^+\rangle$ seen in Eq. (2.6). In Section 3 we shall describe the calibration procedure implemented to achieve the maximally entangled state. All things considered, this geometry generates the entangled Bell state introduced in Section 2.1, and is the configuration characterized experimentally in Section 3.4.

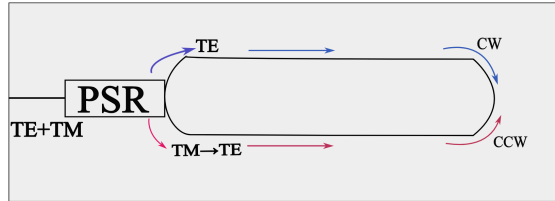


Figure 2: Basic sketch of a Sagnac-loop geometry for polarization-entangled photon-pair generation. In this example, the polarization-selective element splitting the pump into CW and CCW processes is a polarization-splitter rotator.

3 Methods

3.1 Silicon photonic integrated circuit

3.1.1 Chip material platform

The photonic chip (see Figure 3b) was fabricated by Applied Nanotools foundry, using components from the Alcyon Photonics process design kit (PDK) on a silicon-on-insulator (SOI) platform and has dimensions of 1×1 cm. Figure 3a shows a schematic cross-section of the chip, illustrating a silicon rib waveguide above the buried silicon-dioxide (BOX) coat.

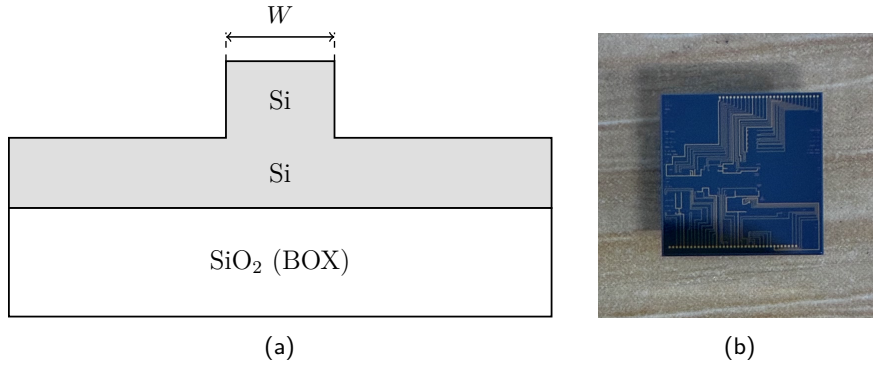


Figure 3: PIC description depicting (a) Cross-sectional schematic of a silicon-on-insulator (SOI) rib waveguide of width W , and (b) a photo of the chip.

3.1.2 Chip layout

In addition to the target polarization EPS, the PIC also features test structures used to characterize its propagation, coupling, and polarization-handling performance.

The first group consists on a set of four waveguides sharing a width of $W = 0.5\mu\text{m}$ in varying lengths, with input and output ports on opposite sides. The chip also features a polarization beam splitter (PBS) and a polarization-splitter rotator (PSR), both of width $0.5\mu\text{m}$, encompassing the full chip's length. Lastly, the PIC contains a set of six nominally identical EPSs, consisting on a PSR followed by a Sagnac loop, tracing the geometry described in Section 2.3.4. They differ only in length and width: three of the sources span a length of 1cm, whilst the other 0.5cm; the waveguide width is varied across $W = 0.5$, 0.55 , and $0.6\mu\text{m}$. Unlike the first set, here the same physical port serves as both input and output.

By testing different combinations of widths and lengths we identify the geometry that optimize the EPS generation efficiency and quality.

3.2 Experimental setup

The characterization of the EPS entails various physically distinct experimental stations. We begin with a classical setup, used to assess the linear and nonlinear optical properties of the chip prior to any quantum light generation. After this, we adapt the scheme so that it allows for quantum photon-pair generation via SFWM, to which a final entanglement certification station will be added –used to evaluate polarization correlations and certify the generated entanglement. Before describing each station individually though, we outline the pump source and chip-coupling and monitoring hardware, common to all.

All measurements are driven by a continuous-wave, tunable laser: the Pure Photonics PPCL550, operating across the C-band, with an output power between 6 dBm and 17 dBm. For all experiments the pump was set at 1550.12 nm, which corresponds to the ITU channel 34¹.

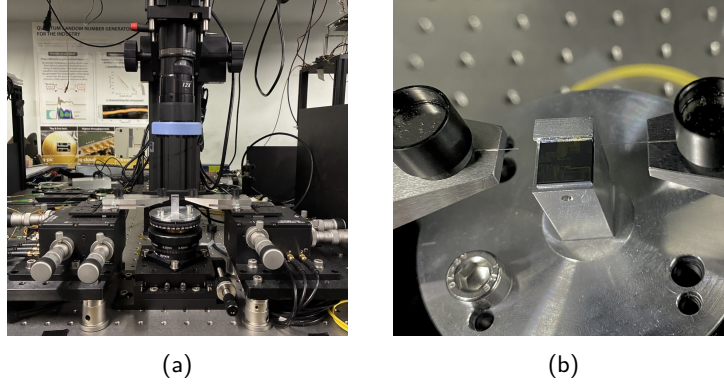


Figure 4: **(a)** Chip mounting scheme. The chip itself sits on a chip holder, which in turn is on a lab jack allowing for coarse vertical positioning (z -axis); the lab jack is mounted on a translation stage used for coarse positioning along the chip's length. Light is coupled on and off-chip by edge coupling, using tapered (lensed) single-mode fibres positioned on either side of the chip, held by a V-groove fibre holder. Said holder is mounted on a Thorlabs NanoMax stage (MAX313D), providing control over three degrees of freedom (x, y, z) for both input and output fibres. As the NanoMax stage allows for control in the micrometer regime, it is fixed on a translation stage for rougher positioning. Finer tuning is available via the 3-axis piezo-controller that each NanoMax stage is connected to. **(b)** Close-up of fibre-to-chip coupling, showing how the tapered fibres are edge-coupled to the facet.

Figure 4 shows how the chip is mounted and coupled. Given that edge coupling requires the tapered fibre tip to be positioned within submicron precision on the waveguide facet, we need to be able to see the coupling region. For that purpose, a 12X infrared (IR) zoom lens is positioned above the chip and connected to a PC for live imaging. An infrared ring light is mounted around the lens to illuminate the chip's surface.

As a final note, given that the fibres are not polarization-maintaining, all fibre segments are either taped to the optical table or secured in a fibre optic tray, to prevent unintended polarization changes due to random phase-shifts induced by mechanical drifts.

3.2.1 Classical characterization

To characterize the test structures described in Section 3.1.2—namely, the set of waveguides of different lengths, PBS, PSR—a simple transmission setup was used. All these structures have their input and output ports on opposite sides of the chip, hence the only adjustment needed is adequate fibre coupling.

The pump laser is routed through a free-space polarization controller consisting of a quarter-wave plate (QWP), followed by a half-wave plate (HWP) and a second QWP, allowing manual adjustment of the input polarization. From the polarization controller (PC) a single-mode fibre connects to the tapered fibre, which couples the pump into the

¹International Telecommunications Union (ITU) channels are standardized optical wavelengths used in Dense Wavelength Division Multiplexing (DWDM), allowing multiple signals to share one fibre.

chip. The beam travels through the device, and exits the PIC through the output tapered fibre, positioned on the other edge of the PIC, and connected to an power meter.

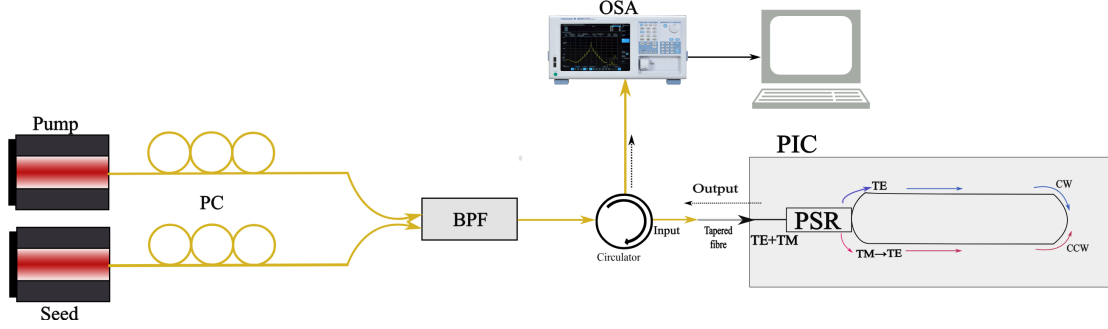


Figure 5: Experimental setup for stimulated four-wave mixing measurements. The pump and tunable seed lasers are independently polarization-controlled and combined before being coupled into the Sagnac-loop device through the input/output PSR. A circulator separates the signal returning from the chip from the incident fields and routes it to an optical spectrum analyser (OSA)

By contrast, in the Sagnac loop structure the PSR serves as both input and output, thus the setup must be modified to accommodate this. To allow for the sFWM measurements, two lasers are combined at the input: the *pump* laser, and a tunable continuous-wave *seed* laser (Santec TSL-570), set to the signal or idler wavelength. As shown in Figure 5, each laser passes through its own free-space fibre-bench PC before the two beams are combined using a band-pass filter (BPF), used in the combiner configuration, centred at 1550.12 nm (i.e. each laser is connected to the pass or reject port, and the mixed signal exits through the combined port). Next, the combined signal is connected to port-1 of a circulator, whose port-2 connects to the input tapered fibre. Upon exiting the chip through the same port, the beam re-enters the circulator at port-2 and exits at port-3, from which it is sent to an optical spectrum analyser (OSA).

3.2.2 Quantum photon-pair generation

For the quantum photon-pair generation, the seed laser and the OSA are removed, and the setup is modified to include the filtering stages required to isolate the generated signal and idler photons from the residual pump, which is many orders of magnitude brighter.

Upon exiting the laser, the pump meets a BPF1 centred at 1550.12 nm, to ensure a clean pump is transmitted. The polarization state is manually set by a PC, and upon exit it meets a variable optical attenuator (VOA), to easily control the power, followed by a 99/1 fibre-splitter. The 1% output is monitored on a power meter to track the input power non-invasively, while the 99% output is connected to port-1 of a fibre circulator, and port-2 connects to another BPF2. Admittedly, BPF2 is not required for SFWM, however it allows to monitor the coupling throughout the experiment: the pass port containing the rejected pump power exits through port-3 and to a power meter, as Figure 6 depicts.

On-chip, the pump enters the PSR, which spatially separates its TE and TM components, and rotates the TM polarization to TE. The two resulting TE-polarized fields propagate clockwise (CW) and counter-clockwise (CCW) simultaneously around the waveguide. Each direction independently drives a SFWM process, annihilating two pump photon to generate a correlated signal-idler photon pair (see Section 2.3.4). Having traversed the full loop, each counter-propagating field now contains residual pump together with newly-generated signal and idler photons, and return to the PSR. There, the field corresponding

to the CCW arm is rotated from TE back to TM, while the CW field remains in TE. The two are recombined at the PSR output: a coherent combination of two indistinguishable, orthogonally-polarized SFWM amplitudes constitutes the polarization-entangled state of Eq. (2.39).

Upon exiting the chip, the beam containing the pump, along with the generated pair, undergoes a cascade of filters to isolate the signal and idler. First, as mentioned, it re-enters BPF2, where the pump is first removed. It is then followed by a notch filter (Opneti, centred at 1550.12 nm) that strongly attenuates the residual pump while transmitting the signal and idler photons. This field is then sent to a first dense wavelength-division multiplexer (DWDM, ITU channel 27), which separates the two: one output port carries the idler, sent directly to a single-photon avalanche diode (SPAD); the other carries the signal along with residual out-of-band pump, passing through a second DWDM stage (channel 41²), whose output is sent to a second SPAD. Both SPADs are connected to a Swabian Instruments Time Tagger for single counts and coincidence analysis.

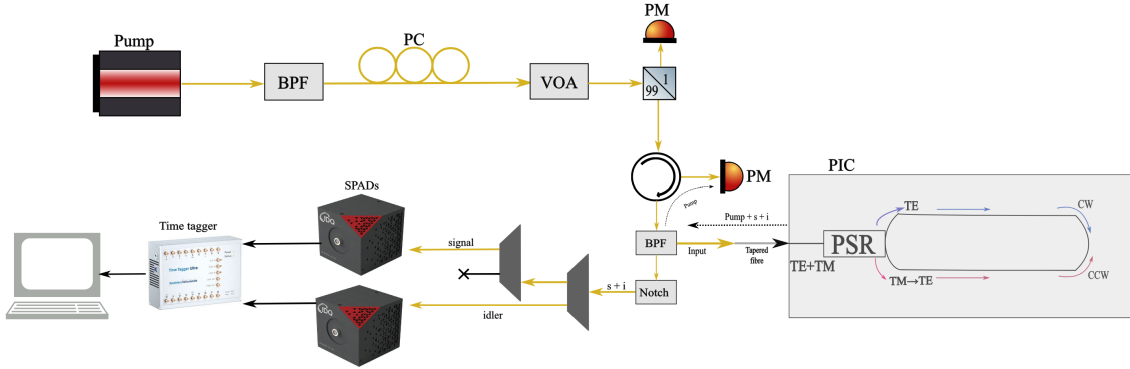


Figure 6: Experimental setup for SFWM. The pump is spectrally filtered, polarization-controlled, attenuated, and coupled into the Sagnac-loop device. The generated signal and idler photons are separated from the residual pump and from each other by a cascade of spectral filters and DWDMs, and are then routed to the SPADs.

3.2.3 Entanglement certification station

The entanglement certification station extends setup from Figure 6: rather than being sent directly to the SPADs, the signal and idler outputs are routed to an independent polarization-analysis station (see Figure 7). We assign the labels Alice and Bob to the signal and idler, respectively.

Each chain begins with a PC. The light is then coupled to a polarization analysis module (PAM), one for Alice and one for Bob. Each PAM consists on a free-space HWP mounted on a motorized rotation stage which, by turning by an angle θ , rotates the incident linear polarization state by 2θ ; followed by a fixed polarizing beam-splitter (PBS) that projects the rotated state onto the H/V basis into two separate output fibres. Thus, setting the HWP angle selects the measurement basis (e.g. H/V at $\theta = 0^\circ$, D/A at $\theta = 22.5^\circ$), while the PBS performs the actual projective measurement.

This way, each PAM yields two output fibres per arm. The four fibres are passed through another PC, before going to the superconducting nanowire single-photon detector (SNSPD). This last PC does not set the measurement basis: it rotates the polarization

²The DWDM channels are chosen symmetric to channel 27, corresponding to the pump's wavelength, as recall that $2\omega_p = \omega_s + \omega_i$.

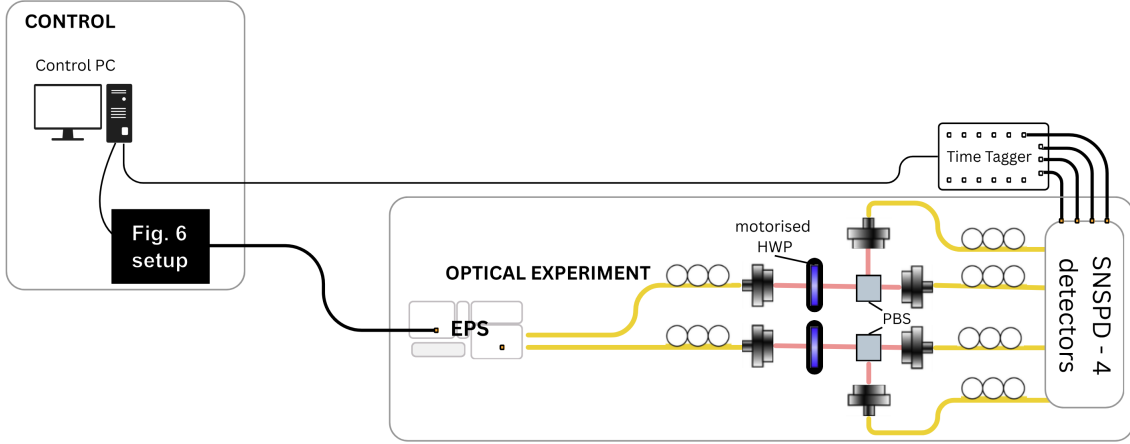


Figure 7: Polarization analyser module scheme. The outputs of the DWDMs are connected to the first set of polarization controllers, prior to the motorised HWP. The black box represents the full setup enabling SFWM generation.

state onto the birefringent axis of the polarization-maintaining fibre of the corresponding SNSPD channel, maximizing detection efficiency. All four SNSPD output channels are connected to the same time tagger as before.

The time tagger is connected to the laptop, from where we can access each channel's single counts and control the rotator motors of the PAMs, all from the same software.

3.3 Single photon detection

In this work two different single photon detectors have been used: single-photon avalanche diodes (SPADs), and superconducting nanowire single-photon detectors (SNSPDs).

A SPAD is a p-n junction reverse-biased slightly above its breakdown voltage. When a single photon is absorbed it triggers an avalanche of charge carriers, producing a macroscopic current pulse. After each detection event the avalanche must be stopped, and the diode bias restored, giving SPADs a relatively long dead time and making them prone to saturation at higher count rates.

On the other hand, an SNSPD uses a thin superconducting nanowire cooled down to cryogenic temperatures. A current bias is passed through the wire, just below its critical current. In this case, photon absorption locally suppresses superconductivity, and the sudden resistance creates a voltage pulse; said pulse is read as the single-photon event. Recovery entails the cooling back down of the nanowire. Because each of the four SNSPD channels used is biased independently, each channel's efficiency can be optimized separately.

Relative to SPADs, SNSPDs offer substantially higher detection efficiency and much shorter dead times, allowing them to sustain higher photon flux without saturating. These advantages motivate their use for the polarization-entanglement measurements, as it is paramount to have unsaturated coincidence statistics across the four channels.

3.4 Experimental procedures

Three nominally identical copies of the chip described in Section 3.1.1 were available for characterization. Propagation loss, coupling loss, and PBS/PSR extinction ratio (Sections 3.4.2, 3.4.3) were measured across several copies, checking device-to-device variability. Based in this characterization, the copy exhibiting the highest PSR extinction ratio was

selected for the subsequent measurements: stimulated and spontaneous FWM, and entanglement certification (Sections 3.4.4–3.4.7).

3.4.1 Fibre-to-chip alignment and coupling

Prior to any measurement, the input and output tapered fibres must be properly aligned to the waveguide facet (as Figure 8 shows). It must be noted that the test structures have different input/output layout compared to the EPS, with different facets involved.

First and foremost, via the lab jack and the translation stage the chip is positioned such that it is visible under the camera, illuminated by the IR light. Each fibre is then brought coarsely into position using, if need be, the translation stage holding it, but mostly via the NanoMax stage’s adjusters. The goal in this initial stage is to bring the fibre tip visually close to the facet, as shown in Figure 8.

At this point, the laser is turned on and the iterative alignment process begins. Starting with the input, the NanoMax stage is scanned along x , y , z while monitoring the transmitted power at the output on a power meter, until a local maximum is found. At which point, the output fibre is scanned the same way. This input-output optimization process is iterated several times, as each new local maximum enables an optimal position for the opposite fibre, until the transmitted power converges to a stable maximum. To further optimize the coupling, we turn to the piezo-controllers (one per each fibre) with three spatial degrees of freedom too, and the same process is carried out until we achieve maximum transmitted power.

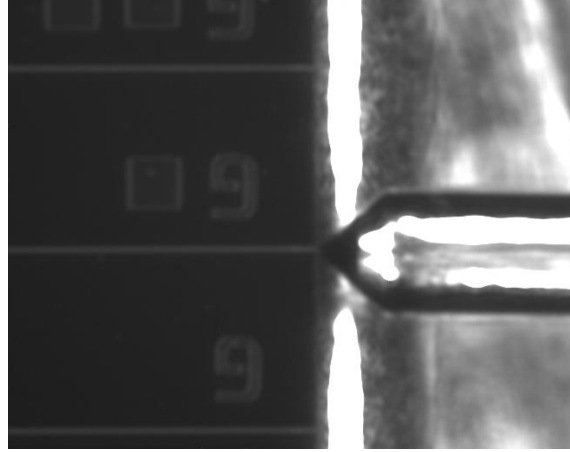


Figure 8: Screenshot of IR camera above the chip, showing a tapered fibre coupled to the waveguide facet.

Given that the coupling is progressively lost over time, in longer measurements such as the entanglement certification, coupling is monitored through the power meter tap described in Section 3.2.3: the circulator port-3 for the Sagnac device. This way, optimal alignment is verified without having to touch the fibres carrying the signal, potentially inducing unwanted random phase-shifts. Overall, when the monitored power drops beyond 0.2 dBm relative to its maximum value realignment is repeated.

3.4.2 Propagation loss and coupling loss

Using the transmission setup depicted in Section 3.2.1, the pump is coupled to each of the four waveguides of varying length, following the coupling procedure described in Sec-

tion 3.4.1. For a fixed input power, the transmitted power is measured for each waveguide length. This procedure is repeated for each of the three chip copies.

It is important to measure the power incident on the chip rather than taking the nominal laser power as reference, as the optical path between the laser and the chip introduces additional losses. Thus, we denominate P_{in} the input power, and P_{out} the transmitted power.

Throughout this work, optical powers are measured in dBm, a logarithmic unit defined with respect to 1 mW,

$$P_{\text{dBm}} = 10 \log \left(\frac{P}{1 \text{mW}} \right) \quad (3.1)$$

. Consequently, a difference between two powers in dBm corresponds to their ratio in linear power units:

$$P_{\text{out,dBm}} - P_{\text{in,dBm}} = 10 \log \left(\frac{P_{\text{out}}}{P_{\text{in}}} \right). \quad (3.2)$$

Assigning y the quantity in Eq. (3.2), and assuming equal coupling losses at input and output facets, the total insertion loss of a waveguide of length L can be expressed as

$$y = -2\alpha_c - \alpha_p L, \quad (3.3)$$

where α_c is the coupling loss per facet (dB/facet), and α_p is the propagation loss (dB/cm). Thus, by performing a linear fit of the obtained experimental results we obtain the desired losses.

3.4.3 PBS and PSR extinction ratio

The same transmission setup described in Section 3.2.1 is used to characterize the polarization extinction ratio of the PBSs and PSRs. After coupling the input fibre to the chip, the output fibre is initially positioned at the facet of one of the two waveguides of the optical device (Figures 10a and 10b). Then, the input power is set to the desired polarization state, using the PC, as shown in Figure 9c.

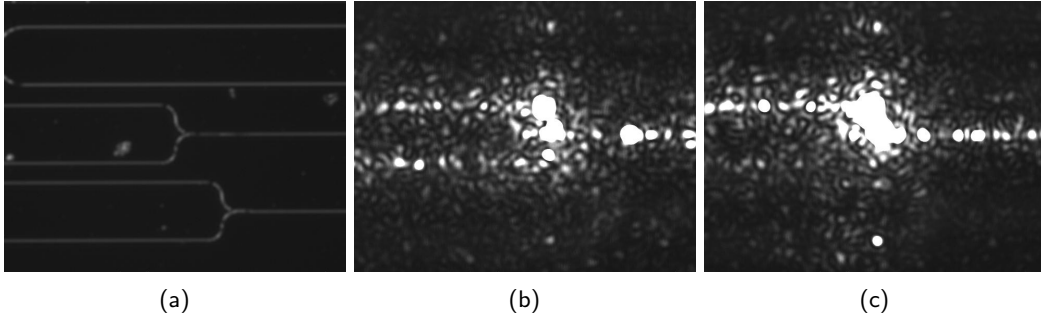


Figure 9: Infrared-camera images of the chip during PBS characterization. **(a)** No laser light is coupled into the waveguide. **(b)** Laser light is coupled into the device with a mixed polarization state. **(c)** The input polarization is adjusted to predominantly excite the TE mode, resulting in most of the transmitted light propagating through the upper waveguide. The residual light visible in the lower waveguide is attributed to the finite polarization extinction of the device.

The same measurement procedure was used for both PBS and PSR characterization. First, the input polarization was adjusted with the PC to maximize the transmitted power in the selected output channel. The corresponding maximum power, P_{max} , was recorded while keeping the fibre coupling fixed. The input polarization was then rotated by 90° using

the half-wave plate (HWP), without modifying the fibre alignment or any other coupling condition. The resulting minimum power, $P_{\min}$, measured at the same output channel was then recorded.

This procedure was repeated for both orthogonal input polarization states and for each relevant output channel. For the PBS, the measurement was performed at each output waveguide to quantify the suppression of the unwanted polarization component. For the PSR, the same procedure was applied at the selected output port to determine the polarization-selective transmission. The extinction ratio was calculated from the ratio between the maximum and minimum transmitted powers,

$$\text{ER} = 10 \log_{10} \left(\frac{P_{\max}}{P_{\min}} \right). \quad (3.4)$$

Here, $P_{\max}$ corresponds to the transmitted power for the desired polarization state, whereas $P_{\min}$ corresponds to the residual power measured after rotating the input polarization to the orthogonal state.

3.4.4 Stimulated four-wave mixing (sFWM)

The aim of this experiment was to obtain the spectral response of each waveguide under both TE and TM excitations, allowing the FWM generation bandwidth to be evaluated. In particular, the spectra were used to identify the waveguide providing the broadest generation bandwidth, in order to have more spectral channels available. Furthermore, characterizing the TE and TM generation separately allows their spectral responses to be compared, providing insight into the waveguide configuration that offers the most indistinguishability.

The sFWM measurements described in this section were carried out on a single chip, selected according to the PSR and PSR ER presented in Section 4.1.2. Using the setup of Figure 5), the pump is fixed at $\lambda_p = 1550.12$ nm and power $P_p = 17$ dBm, whilst the seed power is set at $P_s = 12$ dBm ± 0.1 dBm –for the seed laser the power is controlled through an attenuator, rather than being directly configurable. In this case, the attenuation is set at 1.00 dB.

Prior to any measurements, the polarization was adjusted using the PSR, ensuring the combined beam was launched into the desired TE or TM mode. Once the appropriate state had been established, the fibres were recoupled to the Sagnac device, all without switching off the lasers, thereby preserving the polarization settings. The sFWM characterization was then performed by sweeping the seed wavelength while monitoring the output spectrum with an optical spectrum analyser (OSA). The measured spectrum contains the pump and seed components, along with the generated signal and idler fields arising from the FWM process.

The seed wavelength was swept from 1480 nm to 1625 nm in increments of 1 nm. For each seed wavelength, the OSA records the output spectrum over a span of 160 nm centered at 1550 nm, with a spectral resolution of 0.3 nm. The described wavelength sweep was performed for each of the six Sagnac loop waveguides, and for both TE and TM polarization states.

The wavelength sweep and the spectral acquisition were automated under the same centralized control script that simultaneously configured the seed laser wavelength, triggered the corresponding OSA measurements, and saved the obtained data. The source code is provided in A.

3.4.5 Quantum light generation and CAR characterization

For the subsequent quantum measures the setup implemented was the one depicted in Figure 6. Rather than performing the tests on all six Sagnac waveguides, a single circuit was selected based on its FWM generation bandwidth and the similarity of the TE and TM spectral responses.

Before characterizing spontaneous photon-pair generation, the pump was set to either TE or TM polarization using the PSR and subsequently coupled into the selected Sagnac loop. The TE and TM configurations were characterized separately in order to assess the photon-pair generation performance for each polarization state. In this experiment, no seed field was injected into the waveguide, such that the generated signal and idler photons arise spontaneously through the FWM process.

Both SPAD channels were configured with a detection efficiency of 30%, a sensor temperature of -40°C , and a dead time of $1\text{ }\mu\text{s}$. This sets an upper bound on the count rate a SPAD can register without saturating of $1/\tau_{\text{dead}} = 10^6\text{ counts/s}$. However, approaching this limit causes the detector to undercount the true photon flux, so the input power was set to keep single count rates in the range of 20k–30k counts/s.

For each SPAD channel, the time tagger records timestamped detection events (i.e. *single counts*). By correlating the timestamps of both channels within a fixed time window, *coincidence counts* are obtained. Since photon pairs are emitted simultaneously, true coincidences appear as a pronounced peak in the histogram of signal-idler time delays, showed in Figure 10. By applying a software time offset to one channel the peak is centred at zero. Both the histogram –used for peak visualization and offset calibration– and the data acquisition are controlled through the same centralized GUI.

The CAR was measured as a function of the pump power in order to characterize the quality of the generated photon-pair correlations and to determine the operating regime providing a suitable compromise between pair generation rate and accidental coincidences. The pump power was swept from -20 dBm to -12 dBm using the VOA, leaving the source power untouched to avoid disturbing the polarization. For each power, the integration window was set to 400 ps and the acquisition time to 60 s, measuring single counts and coincidences. During the scan, the coupled power was monitored to ensure that it remained stable throughout the experiment. The measurement was repeated independently for TE and TM polarization, allowing their photon-pair generation performance to be compared.

3.4.6 Basis calibration and reference-state verification

For the upcoming experiments, we transition to the use of the SNSPDs, with the setup described in Section 3.2.3. Prior to any measurement, the bias current of each of the four SNSPD channels must be individually calibrated. For each channel, the bias current was gradually increased while monitoring the corresponding single count rate: at low bias, count rate moves linearly with the bias, reflecting increasing detection efficiency as the nanowire is biased closer to the critical current; beyond a threshold, the count rate increases exponentially, dominated by dark counts. Consequently, each channel’s bias was set slightly below its threshold, maximizing detection efficiency whilst avoiding excessive noise.

Before attempting to characterize the entangled state, the PAMs must be calibrated using a known, separable reference state, following the projective measurements outlined in Section 2.2.3. This preliminary measurement serves to, in one respect, establish the correspondence between the polarization axes of the chip and those defined by the PAMs; and secondly, to balance the four detection channels. In other words, this test is done for a state whose expected polarization correlations are known a priori, it is not an entanglement

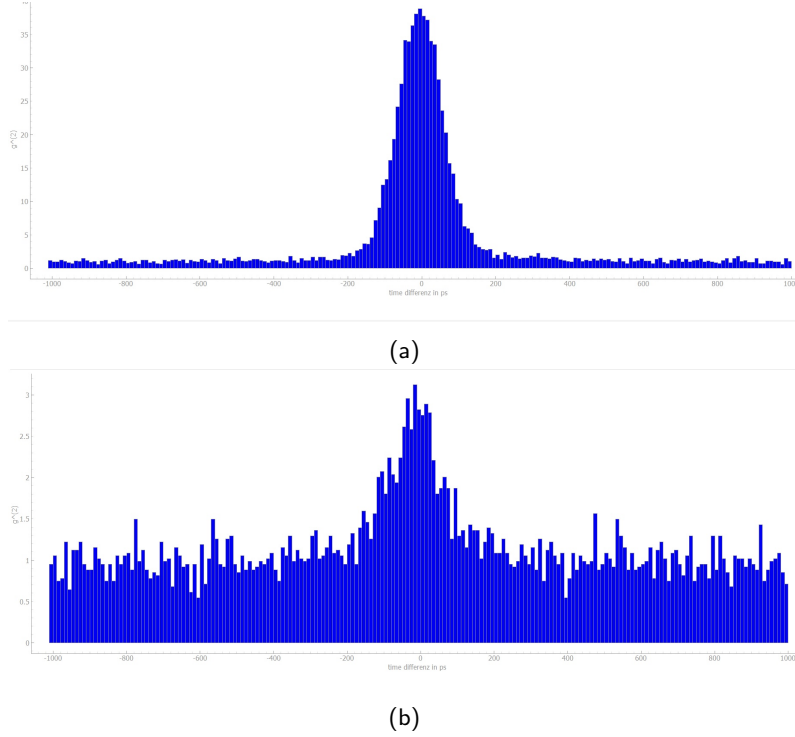


Figure 10: Temporal correlation histograms, plotted as the normalized second-order correlation function, $g_{st}^{(2)}(\tau)$, versus the detection-time difference τ in ps. **(a)** Histogram corresponding to desired coincidences (HH or VV), showing a pronounced peak, well above the dark counts background, centred at $\tau = 0$ after applying the appropriate timing offset. Its width is around 400ps, which will be the integration window, **(b)** shows the cross-term coincidence channel (HV or VH), exhibiting a weak peak, with a magnitude close to the accidental-coincidence background.

measurement.

The reference state is prepared by deliberately pumping only one direction of the Sagnac loop. This is achieved by preparing the pump in a pure polarization state with the PSR, as there won't be any counter-propagating field to yield a superposed state upon recombination.

When preparing the pump in TE-mode, at the output of the loop this corresponds to the state $|H_A H_B\rangle$, and is therefore separable. The subscripts A, B refer to Alice and Bob stations. Similarly, preparing the TM-mode yields $|V_A V_B\rangle$. Thus, we are establishing the TE (TM) polarization to correspond to H (V).

The first stage of the calibration consists on aligning the polarization scheme of the two PAMs with the physical polarization axes of the chip. Both HWP in the PAMs were initially set to the H/V basis. Since the reference state generated under pure-TE pumping is expected to be $|HH\rangle$, the PC2 were tuned so as to maximize the single counts Alice- H /Bob- H , while minimizing coincidences in the three orthogonal combinations, namely $H_A V_B$, $V_A H_B$, and $V_A V_B$.

To compensate for unequal transmission and detection efficiencies among the SNSPD channels, the analyser basis is changed to the diagonal basis, D/A , by rotating the HWP in each PAM from $\theta = 0^\circ$ to $\theta = 22.5^\circ$, thus yielding the state from Eq. (2.28), and the state is equally projected onto the two output ports of the PBS. By adjusting PC3 (which only affects coupling efficiency), the single rates were balanced within each arm: $S_{A,H} \simeq S_{A,V}$,

$$S_{B,H} \simeq S_{B,V}.$$

After completion of both calibration stages, the analyser basis was returned to H/V , and coincidence measurements were then acquired for an integration time of 60 s. The same procedure was repeated in the V/V configuration. Both measurements were performed in both pure-TE and TM pumping, prepared using the PSR. Finally, under TE pumping, the diagonal basis was used to perform the corresponding acquisition too.

This experiment ensures that the ensuing two-photon interference measurements can be interpreted in a calibrated polarization basis, without spurious correlations arising from unequal detector-channel responses.

3.4.7 Two-photon polarization interference

This final experiment aims to characterize the polarization correlations produced when the Sagnac source is under bidirectional pumping. In particular, the target state in this test was $|\Phi^+\rangle$, from Eq. (2.6). Recall that two states with the same components ($|HH\rangle$, $|VV\rangle$ in this case) but different relative phase exhibit different correlations when measured in a rotated basis. For this reason, the target state is prepared while checking its polarization correlations in two complementary bases: H/V and D/A .

Thus, PC1 must be adjusted to maximize the coincidence counts $H_A H_B$ and $V_A V_B$, while minimizing cross-polarized channels. This is done iteratively switching between the complementary bases, until the observed correlation pattern was stable under switching between the two bases. As is illustrated in Section 2.2.3, observing stable correlations in both mutually unbiased bases confirms that the source is operating close to the intended Bell-state configuration, rather than merely producing correlated photons in a single basis.

Once the Bell state configuration has been established, we move on to measuring the angular dependence of the pair's correlations. The idler and signal outputs are labelled as Alice and Bob, respectively. Then, Alice's rotation motor is fixed while Bob's is rotated, therefore provide a quantitative characterization of the polarization correlations of the photon pairs, as is explained in Section 2.2.3.

The fringe scan as performed first with Alice fixed in the H/V basis, at $\theta = 0^\circ$. Bob's HWP was rotated in physical increments of $\theta = 15^\circ$, which corresponds to a change of 30° in the polarization. At each angular setting, and for a full wave rotation, coincidence counts were acquired for an integration time of 60 s and a coincidence window of 400 ps. This scan was repeated with Alice fixed in D/A .

The H/V -basis measurement verifies that the photon pairs are predominantly detected $|HH\rangle$, $|VV\rangle$ components, consistent with $|\Phi\rangle = \frac{1}{\sqrt{2}}(|HH\rangle + e^{i\phi}|VV\rangle)$. On the other hand, the complementary diagonal-basis measurement probes the relative phase. Stable, high-visibility correlations in both mutually unbiased bases therefore proves that the EPS produces the intended coherent Bell state rather than merely classically correlated polarization pairs.

3.5 Figures of merit

In this section, we define the quantities used throughout this work to characterize the performance and quality of the generated photon-pair source. These figures of merit quantify both the rate and quality of the detected photon-pair correlations, as well as the quality of the two-photon interference.

3.5.1 Coincidences and accidental coincidences

When a photon pair is generated through SFWM, the signal and idler photons are ideally detected in their respective channels within a short time interval. A detection event in either channel is referred to as a *single count*, while the detection of events in both channels within a predefined coincidence window is referred to as a *coincidence*. However, not every measured coincidence corresponds to a photon pair generated in the same SFWM event. The total measured coincidences, referred to as *raw coincidences*, can contain both genuine photon-pair coincidences and coincidences arising from uncorrelated detection events.

The latter are referred to as *accidental coincidences*. They can originate from detector dark counts, residual pump light, Raman photons, or photons belonging to independent SFWM generation events. Such events are statistically uncorrelated between the signal and idler channels, but can nevertheless fall within the coincidence window and therefore be registered as a coincidence [48]. The true coincidence rate can consequently be obtained by subtracting the accidental contribution from the raw coincidence rate,

$$R_{\text{true}} = R_{\text{raw}} - R_{\text{acc}}. \quad (3.5)$$

Treating the two channels' singles counts as independent Poissonian processes with rates r_1 and r_2 , the expected *accidental* coincidence rate within a window δt is

$$R_{\text{acc}} = r_1 r_2 \delta t. \quad (3.6)$$

This expression shows that accidental coincidences increase with both the singles rates and the coincidence window. Therefore, reducing uncorrelated background counts and choosing an appropriate coincidence window are important for distinguishing genuine photon-pair correlations from accidental events. The ratio between the true and accidental coincidence rates is quantified by the coincidence-to-accidental ratio (CAR), introduced in Section 3.5.2.

3.5.2 Coincidence-to-accidental ratio (CAR)

The coincidence-to-accidental ratio (CAR) is used to quantify the quality of the temporal correlations between the signal and idler photons generated by a photon-pair source. It compares the rate of coincidences arising from true photon pairs with those resulting from uncorrelated detection events. A high CAR therefore indicates strong photon-pair correlations and a low contribution from accidental coincidences, making it an important parameter for characterizing the performance of a quantum light source.

Experimentally, the CAR is obtained by measuring the temporal correlations between the signal and idler detection events. The time differences between detection events are accumulated in a histogram, where genuine photon-pair detections give rise to a coincidence peak around $\Delta t = 0$. The coincidence window should be sufficiently large to account for the timing jitter of the detectors. Coincidences outside the central peak are considered accidental coincidences, which arise from uncorrelated detection events. The CAR is then calculated as the ratio between the true and accidental coincidence rates [48].

Since the CAR depends on the detection efficiency, timing jitter, coincidence window, losses, and noise, it is a parameter that not only characterizes the source but also depends on the experimental setup. Consequently, the same source measured with different experimental configurations can exhibit different CAR values.

The measured coincidence rate R_{raw} between the signal and idler channels is the sum of a true-pair contribution, arising from photon pairs generated in the same SFWM event,

and the accidental contribution R_{acc} given by Eq. (3.6). The true coincidence rate is hence:

$$R_{\text{true}} = R_{\text{raw}} - R_{\text{acc}}, \quad (3.7)$$

and the CAR is defined as:

$$\text{CAR} = \frac{R_{\text{true}}}{R_{\text{acc}}} = \frac{R_{\text{raw}} - R_{\text{acc}}}{R_{\text{acc}}}. \quad (3.8)$$

The CAR therefore quantifies the relative contribution of genuine photon-pair coincidences with respect to accidental coincidences: a large CAR indicates that the measured coincidences are predominantly associated with photon pairs, whereas a CAR close to zero indicates that accidental coincidences are comparable to or larger than the true-pair contribution.

3.5.3 Visibility

The visibility of an interference fringe quantifies the contrast between its maximum and minimum coincidence rates, and therefore provides a measure of the degree of coherence of the interfering probability amplitudes. In the context of polarization-entangled photon pairs, a high visibility indicates that the different probability amplitudes contributing to the interference remain well defined and indistinguishable, which is necessary for observing strong quantum correlations. Consequently, the visibility provides a convenient figure of merit for assessing the quality of the generated entangled state.

For the two-photon interference fringes of Section 3.4.7, each measured fringe is fit to $C(\theta_B) = a \cos(b\theta_B + c) + d$, from which the visibility is

$$V = \frac{C_{\text{max}} - C_{\text{min}}}{C_{\text{max}} + C_{\text{min}}} = \frac{|a|}{d}. \quad (3.9)$$

As shown in Section 2.2.3 (Eqs. (2.33)–(2.34)), $V \rightarrow 1$ is expected for the ideal state $|\Phi^+\rangle$; a reduced visibility indicates distinguishability between the CW- and CCW-generated amplitudes.

The visibility is also related to the maximum CHSH parameter that can be obtained from the corresponding two-photon interference correlations. For sinusoidal correlations with visibility V , the CHSH parameter is

$$S = 2\sqrt{2}V. \quad (3.10)$$

Since local hidden-variable theories satisfy the CHSH inequality $S \leq 2$, a visibility satisfying

$$V > \frac{1}{\sqrt{2}} \approx 70.71\% \quad (3.11)$$

corresponds to $S > 2$ and therefore indicates that the correlations are sufficient to violate the CHSH inequality. Thus, the measured fringe visibility provides a direct figure of merit for determining whether the observed two-photon interference is compatible with CHSH-violating correlations.

4 Results

4.1 Classical characterization

4.1.1 Propagation and coupling loss

Figure 11 shows the measured insertion loss as a function of waveguide length for both chips, together with the corresponding linear fits (Eq. (3.3)). The fitted propagation loss and coupling loss, α_p and α_c , respectively, are reported in Table 1 for each chip.

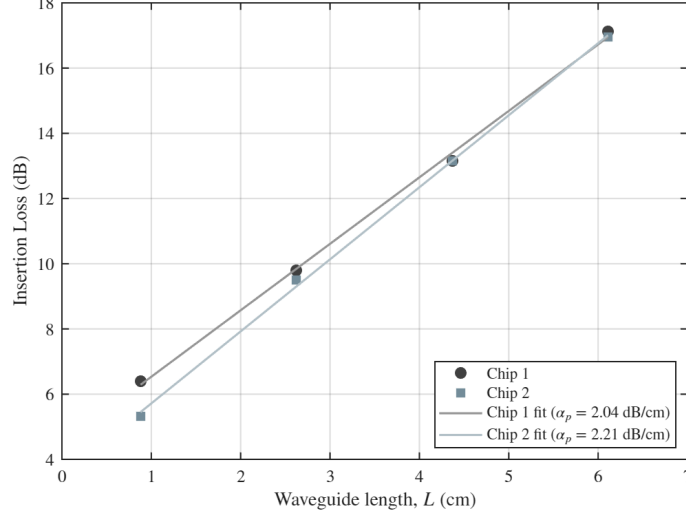


Figure 11: Insertion loss versus waveguide length for Chip 1 and Chip 2, with linear fits used to extract propagation and coupling loss.

	Chip 1	Chip 2
α_p	2.0384	2.2096
α_c	2.2476	1.7528

Table 1: Fitted propagation loss α_p (dB/cm) and coupling loss α_c (dB/facet) for Chip 1 and Chip 2.

Both chips show closely matched performance, with α_p differing by less than 0.2 dB/cm and α_c by less than 0.5 dB/facet between copies, as expected from the fabricated reproducibility across the same wafer.

Chip 3, the copy ultimately selected for the subsequent tests, was not characterized for propagation and coupling loss, as this measurement was identified prior to identifying it as the best-performing device. Given the close agreement between Chip 1 and Chip 2, it is reasonable to assume Chip 3's propagation and coupling losses to fall within the same range.

4.1.2 PBS and PSR extinction ratio

The measured extinction ratios (ERs) of the PBS and PSR test structures are reported in table 2 for all three copies. Chip 1 was discarded from further consideration early in this comparison, as the PBS was extremely asymmetric and the PSR appeared to be defective.

While Chip 3's PSR performed substantially better than its counterparts, what caused it to be ultimately selected for the subsequent tests was the near-symmetric performance

	Chip 1		Chip 2		Chip 3	
	TE	TM	TE	TM	TE	TM
PBS	23.40	3.44	13.90	6.57	12.23	12.30
PSR	6.70	–	13.75	25.00	12.05	29.94

Table 2: Measured polarization extinction ratios (ER) of the PBS and PSR for the three characterized chips, in dB.

of the PBS. A balanced PBS is desirable in the context of FWM, as it enables comparable transmission and collection efficiencies for both polarization channels, reducing polarization-dependent bias.

4.2 Stimulated four-wave mixing

Stimulated four-wave mixing was used to select the waveguide geometry best suited for the succeeding SFWM experiment. Following the procedure outlined in Section 3.4.4, the pump was fixed at $\lambda_p = 1550.12$ nm while the seed wavelength was swept from 1480 nm to 1625 nm. For each seed wavelength, the OSA recorded the full output spectrum, in which the pump, seed, and the newly generated signal and idler peaks are simultaneously visible.

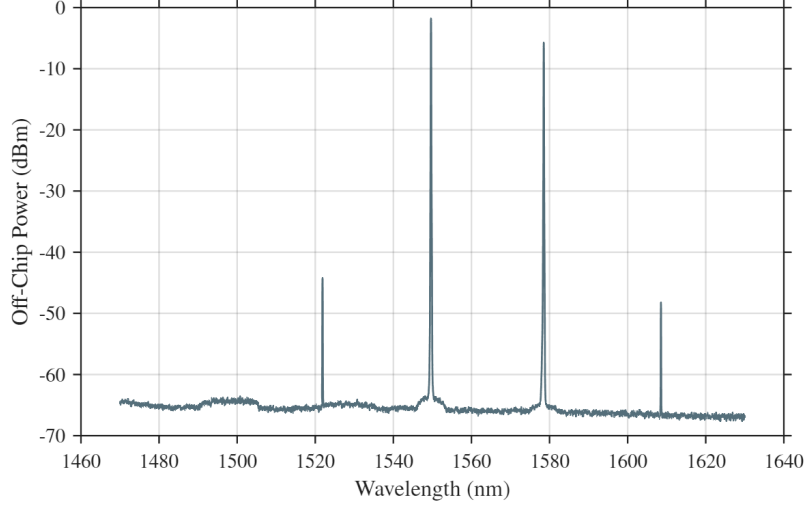
Figure 12a shows an example of OSA trace for a single seed wavelength. Upon full band scan, it is observed that the intensity of the generated signal peak depends on the seed wavelength. Moreover, there is generated signal in the full swept spectrum, thus yielding a large bandwidth, suggesting a broadband source: 1480 nm–1625 nm.

Repeating the wavelength sweep for both TE and TM pump polarization allowed for the normalized conversion efficiency,

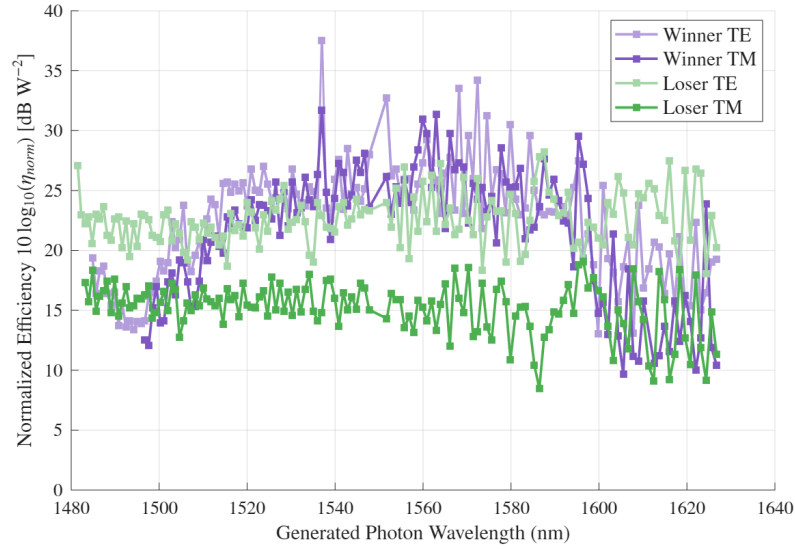
$$\eta = \frac{P_{\text{idler}}}{P_{\text{seed}} P_{\text{pump}}^2} \quad [\text{W}^{-2}], \quad (4.1)$$

to be extracted as a function of the generated-photon wavelength for each device. Since the target application (Section 2.3.4) relies on the coherent combination of two counter-propagating SFWM processes, the indistinguishability of the resulting entangled state depends on the two processes generating photons with matched efficiency across the relevant bandwidth. For that purpose, a bandwidth in which the TE and TM-pumped efficiency curves overlap closely is preferable to one in which they diverge, even if the latter shows higher peak efficiency in one polarization.

Figure 12b compares the normalized efficiency of the selected waveguide (length 1 cm, width $0.55 \mu\text{m}$) against the second-best candidate (length 0.5 cm, width $0.55 \mu\text{m}$). In the selected Sagnac, both the curves of both modes overlap across the measured bandwidth: 1485 nm – 1625 nm, though, as can be observed, towards the extremes the difference is significant. By contrast, the other waveguide shows evident mismatch between modes, despite the TM-mode showing higher efficiency.



(a)



(b)

Figure 12: **(a)** Stimulated FWM for the TM mode. The waveguide width is $0.55 \mu\text{m}$ and the length is 1 cm. The seed is at 1578.52 nm. The signal is generated at 1521.84 nm, the idler at 1608.52 nm. **(b)** Normalized sFWM conversion efficiency for two different waveguide settings. The curves for both TE and TM pump polarizations are displayed as a function of the generated photon wavelength.

The conversion efficiency plots show a non-smooth curve, which can be justified by the Fabry-Perot interference occurring at the waveguide. Figure 13 shows the seed off-chip power as a function of seed wavelength (which recall, is constant at input for all wavelengths), exhibiting the ripple behaviour consistent with the Fabry-Perot effect.

4.3 Quantum light generation

Coincidence rates were measured as a function of pump power for both TE and TM pump polarizations, from which the coincidence-to-accidental ratio (CAR) was extracted, following Eq. (3.8).

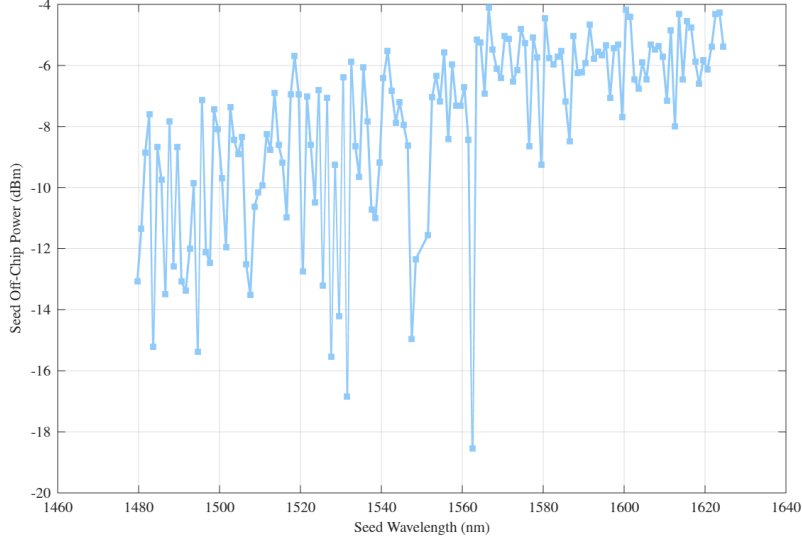


Figure 13: Seed off-chip power as a function of seed wavelength, showing a periodic ripple attributed to Fabry-Perot interference at the waveguide.

Figure 14a shows the measured true and accidental coincidence rates as a function of pump power. Low pump powers correspond to the noise-dominated regime, where we notice how R_{true} and R_{acc} coincide for both polarizations, indicating that the accidental rate is dominated by a background contribution common to both. As pump power increases, true coincidence rate grows with SFWM pair-generation probability, departing from the R_{acc} ; in this regime, genuine pair-generation dominates over background. However, notice how at around 0 dBm the TE and TM curves begin diverging from each other, which can stem from the difference in SFWM conversion efficiency seen in Section 4.2.

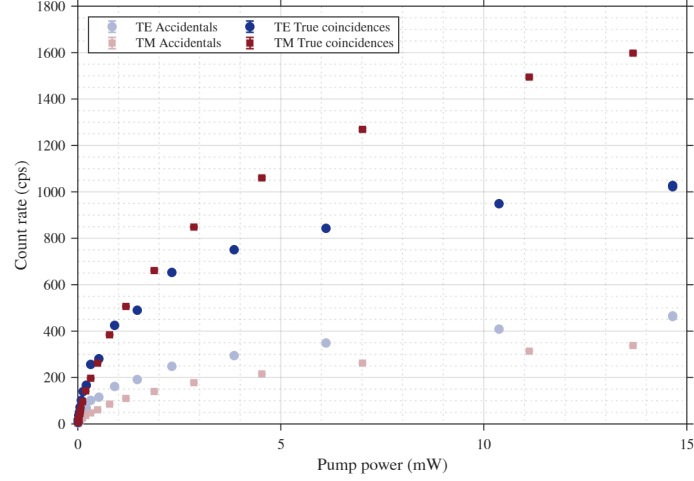
The corresponding CAR as a function of pump power is shown in Figure 14b. The resulting curve is as expected: at low power R_{acc} is limited by a roughly constant background, while R_{true} grows with the pump power, increasing CAR; at higher powers, multi-pair events increasingly contribute to R_{acc} , which grows quadratically with pump power and eventually drives CAR back down.

It is observed how the TM curve lies above the TE curve by roughly a factor of two, whilst the two plots remain coarsely parallel in shape. Given that the true and accidental rates for TE and TM coincide at low power (Figure 14a) and only diverge once pair generation begins to dominate, the CAR offset is more likely attributable to a difference in pair-generation efficiency between the two polarizations than to a polarization-dependent background. This would be consistent with the any asymmetry in the TE/TM conversion efficiency observed in the sFWM characterization (Section 4.2).

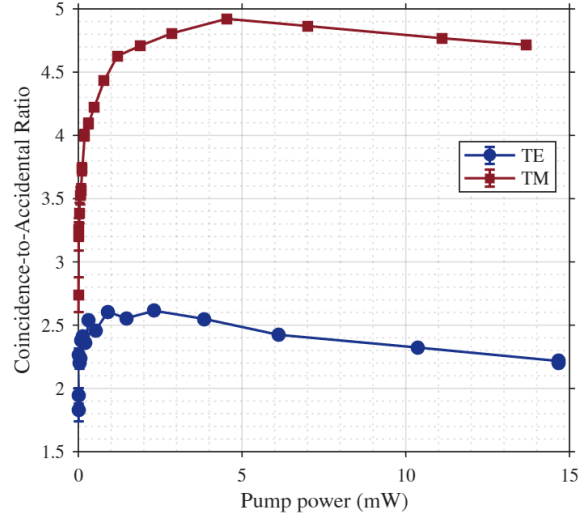
4.3.1 Basis calibration and reference-state verification

As mentioned in Section 3.4.6, prior to the interference measurements, the polarization correlations of the source were characterized in both the rectilinear (H/V) and diagonal (D/A) bases, for both TE and TM pump polarizations. For each configuration, the PAMs were set to a nominal projection and the resulting four coincidence combinations (HH, VV, HV, VH) were acquired and normalized, yielding the coincidence fractions shown in Figure 15.

In the rectilinear basis, the coincidence fractions are strongly dominated by a single outcome for both pump polarizations, confirming that the source produces highly corre-



(a)



(b)

Figure 14: **(a)** Quantum light generation versus pump power. True and accidental coincidence rates as a function of pump power, for TE and TM pump polarizations. **(b)** Coincidence-to-accidental ratio (CAR) as a function of pump power, for TE and TM pump polarizations.

lated photon pairs in the computational basis. Consistent with the polarization rotation imparted by the PSR at the Sagnac loop, the swapped labeling between TE and TM is to be expected.

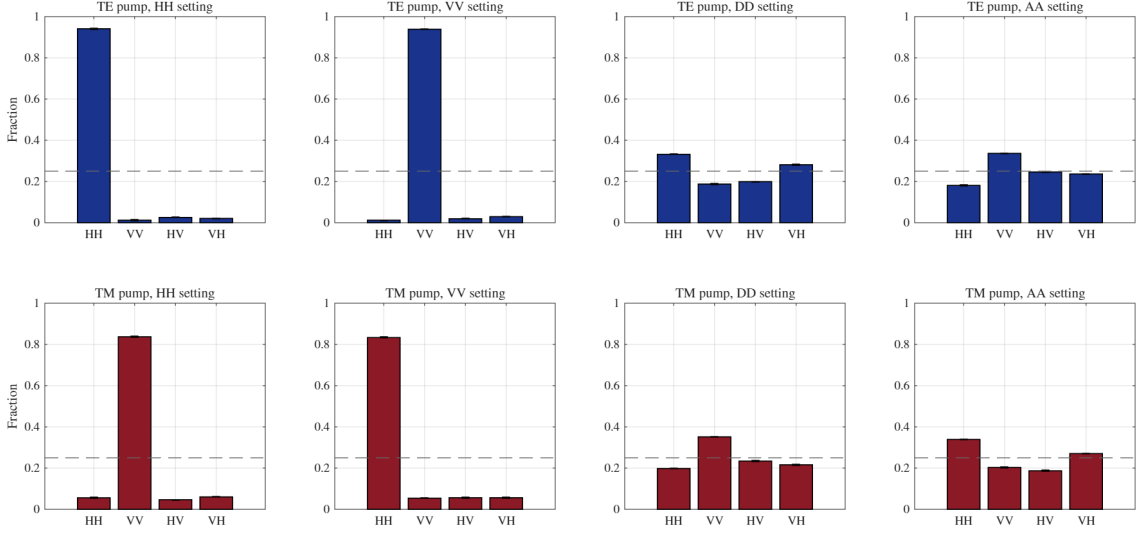


Figure 15: Basis calibration normalized coincidence fractions per configuration.

In the diagonal basis, the dominant outcome mirrors that of the corresponding rectilinear-basis behaviour (i.e. TE pump in DD setting is HH-dominant, as is in HH settings). However, the fractions in this basis are close to the target value of 0.25 (as Section 2.2.3 outlines) for all combinations. A departure from the expected flat distribution cannot be attributed to polarization impurity, since depolarization leaves the diagonal-basis outcome unbiased.

A possible explanation would be a coherent angular offset in the chain (perhaps the analyser axis or the HWP zero point). If it were due to detector efficiency asymmetry, then regardless of which state was pumped, the same physical channel would dominate, which is far from the case.

4.3.2 Two-photon interference fringes

To quantify the quality of the entangled state, the two-photon interference visibility was extracted from the measured coincidence fringes. True coincidences were then fitted to a harmonic model, $C(\delta_B) = a \cos(b\delta_B + c) + d$, thus minimizing the impact of statistical fluctuations. The ensuing plot is presented in Figure 16. With this model, the visibility is determined by the amplitude-to-offset ratio, $V = |a|/d$, as presented in Section 3.5.3. Given that a single measurement basis yields four distinct coincidence curves, the overall visibility for a given basis was computed as the arithmetic mean of each curve's visibility. With this approach, the overall interference visibility was 84.13% for the rectilinear basis ($\delta_A = 0^\circ, 90^\circ$) and 88.65% for the diagonal basis ($\delta_A = 45^\circ, -45^\circ$).

Both basis show visibilities well above the 70.7% threshold required for CHSH violation (Section 3.5.3), confirming genuine polarization entanglement, rather than a classical correlated mixture.

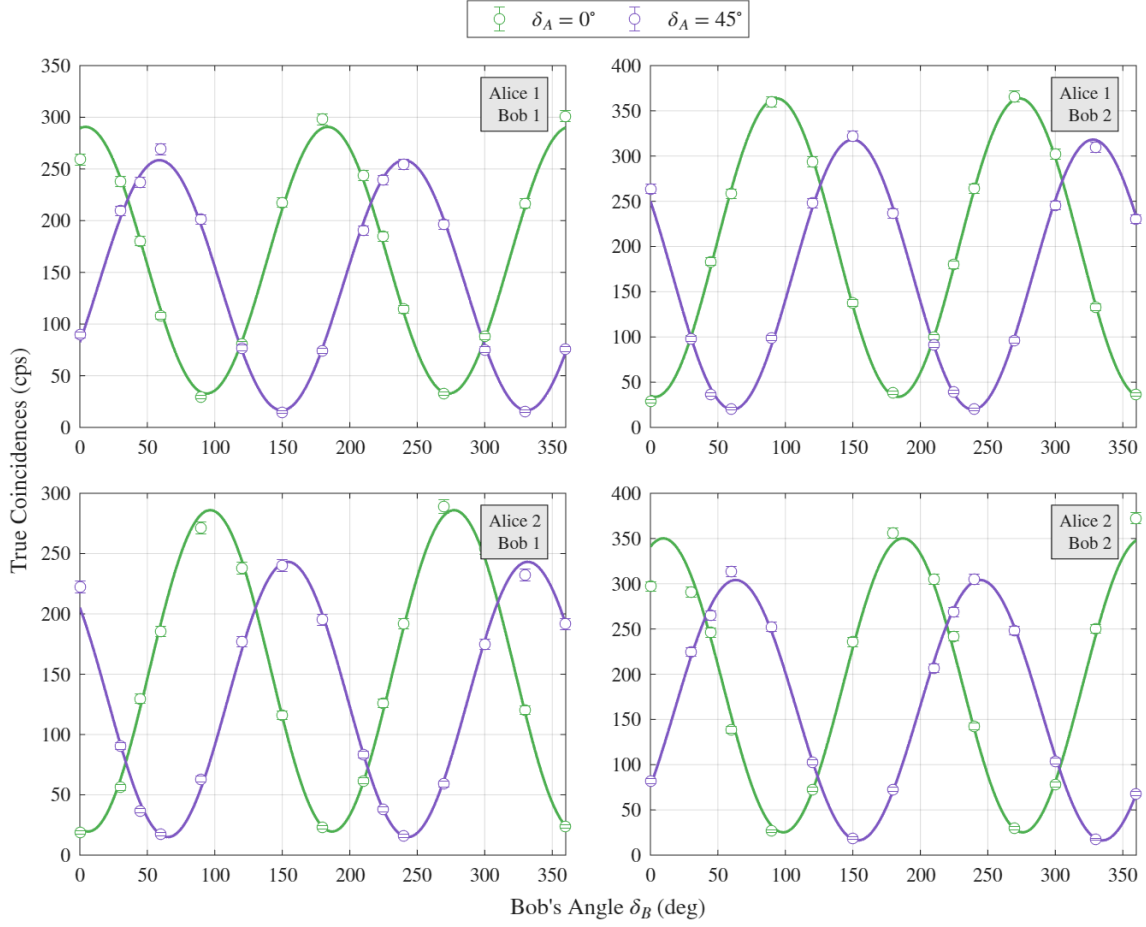


Figure 16: **Two-photon interference fringes.** Coincidence pair rates between Alice and Bob as a function of Bob's polarization angle δ_B . The green and purple datasets correspond to Alice's basis fixed at $\delta_A = 0^\circ$ (rectilinear) and $\delta_A = 45^\circ$ (diagonal), respectively. Measurement settings 1 and 2 (indicated in the gray boxes) correspond to the transmitted and reflected outputs of the respective polarizing beam splitters. Consequently, Alice 1 (2) denotes the detection of the 0° (90°) state for the blue curves, and the 45° (-45°) state for the orange curves. Bob 1 (2) denotes the detection of the δ_B ($\delta_B + 90^\circ$) state. Circles represent the measured true coincidences with Poissonian standard error bars (smaller than marker size); solid lines represent the cosine fitting functions

It is observed how the rectilinear visibility sits lower than its diagonal counterpart. Moreover, in Figure 16 it is further observed that, in all four plots, the $\delta_A = 0^\circ$ (rectilinear, green) fringe consistently reached a higher coincidence rate than $\delta_A = 45^\circ$ (diagonal, purple). This could be justified recalling the asymmetric generation efficiency between the two counter-propagation SFWM processes in the Saganc loop, consistent with the difference in CAR for the two modes seen in Section 4.2. Such imbalance would produce a state closer to $\alpha|HH\rangle + \beta|VV\rangle$ with $|\alpha| \neq |\beta|$, rather than the ideal Bell state (Eq. (2.30)). When Alice is projected onto the rectilinear basis, the resulting curve is dominated by the larger amplitude (either $|HH\rangle$ or $|VV\rangle$), reaching a higher peak rate. By contrast, projecting onto the diagonal basis instead mixes both terms into every outcome, averaging over the stronger and weaker counts, and yielding a comparatively lower rate. Hence, the asymmetry between the two fringes could be a consequence of the TE/TM generation imbalance already noted in Section 4.2.

The fitting parameter c provides the angular position of the fringe peak in each basis. For the rectilinear basis fringe, the peak falls close to the expected 0° reference. However, for the diagonal basis, the peak is found at 54.81° , rather than the nominally expected 45° . This outcome is consistent with the coherent angular offset presumed in Section 4.3.1, where the separable state measurement already deviated from the ideal diagonal setting.

4.3.3 CHSH inequality violation

As a final entanglement certification, the CHSH S parameter was computed from the coincidence data. Recovering the correlation definition presented in Eq. (2.15), for a fixed pair of analyzer settings (θ_A, θ_B) , the polarization correlation function is defined as:

$$E(\theta_A, \theta_B) = \frac{C(\theta_A, \theta_B) + C(\theta_A^\perp, \theta_B^\perp) - C(\theta_A, \theta_B^\perp) - C(\theta_A^\perp, \theta_B)}{C(\theta_A, \theta_B) + C(\theta_A^\perp, \theta_B^\perp) + C(\theta_A, \theta_B^\perp) + C(\theta_A^\perp, \theta_B)}, \quad (4.2)$$

where $C(\theta_A, \theta_B)$ denotes the true coincidence rate (i.e. accidentals subtracted rate, following Eq. (3.7)) between Alice's and Bob's outputs, and $\theta_A^\perp, \theta_B^\perp$ indicate the orthogonal projection at each station. The same sinusoidal fit used to extract the fringe visibility (Section 4.3.2) is now used to evaluate $E(\theta_A, \theta_B)$ at arbitrary θ_B , beyond the discrete angles experimentally measured. The CHSH parameter is thereby

$$S(\theta_B, \theta'_B) = E(0^\circ, \theta_B) - E(0^\circ, \theta'_B) + E(45^\circ, \theta_B) + E(45^\circ, \theta'_B), \quad (4.3)$$

with θ_B, θ'_B set such that they maximize the observed violation. Full implementation details, including the accidentals subtraction, the weighted cosine fits, and the error propagation, are given in Appendix B.

Given the results presented in Section 4.3.2, two sets of (θ_B, θ'_B) were considered. Firstly, following CHSH theory (Section 2.1.5), for a maximally entangled state with equal visibility in both bases, S is maximized when Bob's settings are paired symmetrically about the bisector of Alice's settings: $\theta_B = 22.5^\circ, \theta'_B = 67.5^\circ$. This choice assumes that the D/A correlation peaks at 45° from the H/V basis, yielding:

$$S_{22.5^\circ/67.5^\circ} = 2.22 \pm 0.0078, \quad (4.4)$$

violating the classical bound of $S = 2$ by 28.6σ .

However, as shown in Section 4.3.2, the experimental interference curves for θ_A and θ'_A are actually offset by 54.81° , rather than the nominally expected 45° , thus shifting the angle at which S is maximized. Consequently, we defined a bisector angle β such that $b, b' = \beta \pm 22.5$, and performed a constrained bisector search (preserving the 45° separation between θ_B and θ'_B). This converged to an optimal bisector of $\beta = 146.0^\circ$ ($\theta_B = 123.5^\circ, \theta'_B = 168.5^\circ$), giving:

$$S_{\text{bisector}} = 2.41 \pm 0.008, \quad (4.5)$$

a violation by 51.9σ . A fully unconstrained search (without imposing the 45° separation) was also carried out, the details of which can be consulted in Appendix B, converging to the same result as the constrained bisector.

Overall, both angle choices yield an unconditional violation of the CHSH inequality, thereby certifying that the on-chip source produces polarization-entangled photon pairs.

5 Conclusions and discussions

In this work, we characterized a novel monolithic polarization-entangled photon-pair source on a SOI platform, based on on-chip spontaneous four-wave mixing (SFWM) in a common-path Sagnac loop geometry with a polarization splitter-rotator (PSR). For that purpose, we constructed a probe-station capable of both classical and quantum optical characterization. For starters, we characterized the chip’s propagation loss, coupling loss, and polarization extinction ratios across fabricated copies, selecting the best-performing device. We used stimulated four-wave mixing to identify the waveguide geometry offering the closest TE- and TM-pumped conversion efficiencies, which was subsequently used for generating and characterizing quantum light via SFWM. Finally, the polarization entanglement was certified through two-photon interference fringes, providing an experimental violation of the CHSH inequality. Regarding this last result, the maximum S parameter experimentally measured was of $S = 2.41 \pm 0.008$, exceeding the classical bound, and constituting definite certification that the presented source produces genuine polarization-entangled photon pairs.

Several aspects ought to be addressed to further improve the experiments’ performance. For starters, the measured coincidence-to-accidental ratio showed divergence between the TE and TM pump polarizations, possibly explained by an asymmetry in the pair-generation rate between the two counter-propagating processes; further testing is needed to confirm this, and to properly address it. Moreover, future CAR characterization should be carried out with the SNSPDs, allowing for a more direct comparison between the source’s photon-pair statistics and the detectors ultimately used to certify entanglement. Secondly, and arguably more importantly, the interference fringe offset was found to be roughly 55° , rather than the nominal 45° ; We postulate that with an improved calibration of the half-wave plate (HWP) in the polarization analysis module it is possible to achieve the expected offset angle. Finally, the interference visibility differed between the rectilinear and diagonal bases (84.1% versus 88.7%), a discrepancy worth addressing. We believe that tackling these points would likely improve the achievable S parameter.

Beyond these immediate refinements, the presented source opens the door to several directions for extending this work. For instance, performing a quantum state tomography would provide a fidelity estimate, complementing the figures of merit already obtained, and granting a more complete characterization of the density matrix. Moreover, in the near term, the robustness of the certified entanglement could be assessed in an emulated free-space channel, introducing atmospheric turbulence, attenuation and atmospheric transmittance; emulating a satellite downlink scenario. In the context of free-space QKD, polarization encoding is particularly well suited for atmospheric propagation, and a compact source such as the one presented hither is directly relevant to said application. Looking further ahead, implementing this source in an actual satellite would entail its own dedicated development and testing campaign –packaging, thermal and mechanical qualification, for instance–, as the compactness and CMOS compatibility presented in this work are precisely the attributes that enable the path towards practical satellite quantum key distribution (QKD) architectures. Nevertheless, further development will be necessary to achieve this goal.

More broadly, the device presented in this thesis represents a stepping stone towards bringing high-quality entangled-photon sources onto a scalable, foundry-compatible platform, currently an extremely active field of research. The combination of a common-path Sagnac interferometer with a PSR, both integrated on-chip, removes the need for off-chip polarization control and interferometric stabilization, which have hitherto limited the prac-

ticality of integrated polarization-entanglement sources. This is a significant step for silicon photonic integrated circuits, demonstrating their potential to deliver low-cost and mass-manufacturable hardware required for future quantum communication networks.

We note that the device presented in this work may have potential beyond quantum communication, as entangled photon-pairs are a demanded resource also for photonic quantum computation and simulation. In these fields, on-chip generation is also motivated by a need of scalable, low-loss integration with other circuit elements. A broader question remains as to how far integrated silicon sources can close the performance gap with $\chi^{(2)}$ platforms such as thin-film lithium niobate, whilst retaining their fabrication edge; determining this issue will ultimately determine the deployed quantum-photonic technology.

All things considered, despite the mentioned aspects of the presented source's measured performance show room for refinement, the results obtained in this work stage this device as a robust and promising step towards compact, deployable polarization entanglement sources, be it for satellite-based QKD or, more generally, for the integration of quantum photonics onto scalable and manufacturable technology.

6 Sustainability report and ethical implications

This last section aims to assess the impact and viability of this thesis from a sustainability standpoint. We shall cover its environmental, economic, and social impact, along with the possible ethical implications of completing this Bachelor project and the repercussions of implementing the device on a large scale. Finally, we outline the project’s relationship with the Sustainable Development Goals, or the *2030 Agenda*, proposed by the United Nations.

6.1 Sustainability matrix

The device presented in this work opens the door for many use-cases: from quantum communication to quantum computation and sensing. As each particular application would entail significantly different frameworks, we shall solely focus on the use of this chip on secure communications.

For that purpose, we shall analyse the impact that a mass-produced version of the presented source would have if it were deployed as part of a real quantum communication network.

6.1.1 Environmental impact

The environmental footprint of this work shall be separated into firstly, the actual impact of carrying out this thesis and, secondly, theorizing the repercussions of a mass-produced version of the presented chip.

To begin with, prior to any experimental test, a couple of weeks were invested in familiarizing with the thesis topic, which entailed a lot of literature review. In this initial stage, the work consisted on reading and understanding the materials provided by the supervisor. Notably, an effort was made to avoid printing any paper, so all the reading was carried out on a laptop screen. However, when presented with a concept outside of my comprehension, AI was occasionally used as an aid.

During the hands-on development of the TFG, the energy expenditures fell mostly on the laboratory work, though a smaller fraction also fell on the data processing in MATLAB, yielding additional computational and thus energetic costs.

Overall, the dominant environmental cost was electricity consumption rather than material use. The laboratory laptop controlling the whole setup (from the camera, to laser remote control, to detector’s settings and power) was left powered on continuously throughout all the thesis, independently of whether measurements were actively being taken. The other devices, such as the pump laser, single-photon detectors, the power meter, the camera, and the time tagger were powered only when experiments were being carried out. Though the laptop usage can be justified as being common practice in optics laboratories, motivated by the time cost of setting every device up upon powering it on, it nonetheless represents an avoidable, or at least reducible, source of energy consumption.

The presented factors are recurrent all throughout the project, and are thus used to compute the carbon footprint impact presented in Table 3.

Table 3: Estimated carbon footprint of thesis-related electricity and mobility use.

Source	Used time	Energy consumption (calculation)	Total energy	CO ₂ e
Laptop ^a	700 h	Electricity: $700 \text{ h} \times 90 \text{ W} = 63.0 \text{ kWh}$ Embodied: $700 \text{ h} \times 4.11 \text{ gCO}_2\text{e/h} = 2.88 \text{ kg}$	63.0 kWh	24.93 kg
Pump laser ^b	1080 h	$108 \text{ days} \times 10 \text{ h/day} \times 25 \text{ W}$	27.0 kWh	9.45 kg
SNSPD cryostat ^c	176 h	$22 \text{ days} \times 8 \text{ h/day} \times 1.5 \text{ kW}$	264.0 kWh	92.40 kg
Emails ^d	15 emails	$15 \times 4 \text{ gCO}_2\text{e/email}$	—	0.06 kg
Video calls ^e	10 calls	$10 \times 0.75 \text{ h} \times 200 \text{ gCO}_2\text{e/h}$	—	1.50 kg
General lab electricity	108 days	$108 \text{ days} \times 15 \text{ kWh/day}$	1620.0 kWh	567.00 kg
Train (FGC) ^f	80 min/day	$108 \text{ days} \times 40 \text{ km} \times 14 \text{ gCO}_2\text{e/km}$	—	60.48 kg
Bus ^g	100 min/day	$108 \text{ days} \times 40 \text{ km} \times 100 \text{ gCO}_2\text{e/km}$	—	432.00 kg
Total				1187.82 kg

^a Assuming 180 kg CO₂e / 5-year life.

^b Laser power assumed 25 W ((Pure Photonics PPCL550-class compact tunable module).

^c Cooler input power 1.5 kW.

^d Emission factor 4 gCO₂e per email without attachment.

^e Assumed 45 min average call duration and 200 gCO₂e/h for a small (1–2 participant) videoconference.

^f FGC electric-rail emission factor assumed 14 gCO₂e/passenger-km. Implied average speed ($\sim 30 \text{ km/h}$) is consistent with the stated 80 min/day given stops and urban sections.

^g Assuming 100 gCO₂e/passenger-km.

A second source of consumption, though not present for the most part of the thesis, was the entanglement certification station: operating the superconducting nanowire single-photon detectors (SNSPDs). Operating such devices required maintaining an internal temperature of less than 4 K (cryogenics regime), which is achieved via a continuous input power of the order of 1.5kW, for the full duration of the measurement. This is a considerable load compared to the single photon avalanche diode (SPAD) single-photon detectors, which operated at room temperature. Thus, the carbon footprint of the presented device can be estimated as depicted in Table 3

Beyond electricity, the material footprint of the TFG itself was low: no chip, fibre, or optical component was consumed or discarded during the characterization process, and all hardware (chip, mounts, lasers, detectors, fibres, filters) is reusable shared laboratory components, rather than single-use material.

Considering instead the mass-produced version of this source deployed as a node in a quantum communication network, the choice of platform argued for in Section 1 has direct environmental consequences. Recall that the source is fabricated on a standard silicon-on-insulator (SOI) platform using standard CMOS-compatible foundry process, hence scaling up production does not require building new, dedicated fabrication infrastructure. This is not the case for competing platforms such as TFLN or AlGaAs, whose foundry processes are far less mature and deployed. This fact has direct environmental and economical consequences (presented in the following section), since new fabrication lines entail a major footprint of their own.

It would be an oversight, however, to state that silicon is the cleanest choice. Silicon is indeed highly abundant, but the process required to reach the purity needed for photonic wafers is very energetically costly. By contrast, AlGaAs relies on gallium, an element representing only 0.0015% of the Earth’s crust; moreover, gallium arsenide (GaAs) generates

toxic waste. On the other hand, lithium niobate depends on lithium sourcing, whose environmental costs are well documented at the scale used for battery manufacturing, though the wafer production represents around 1% of the use-cases. All things considered, it is apparent how none of the three competing platforms is inherently desirable, however SOI does exhibit the advantages previously presented.

Another environmental advantage is that the source generates entangled photons at telecom wavelength. Hence, a network built on this source could reuse the existing fibre-optic and DWDM infrastructure, so no dedicated framework would be constructed. Moreover, the photon generation occurs at room temperature, so no active cooling would be needed, in contrast with the classical electronics, where losses in copper force active thermal monitoring (which entails large quantities of water consumption). The main environmental cost would instead sit at the detector stage, assuming it were based on SNSPDs, due to the cryogenics involved, as already mentioned. This can be avoided, trading-off efficiency, by using SPADs instead.

Regarding the limitations, it must be stressed that no data exists yet on the end-of-life impact of such a platform at deployment scale.

From the information depicted in Table 3, we can calculate the estimated impact of this thesis in more tangible magnitudes: trees. We shall consider that 1 elm absorbs 720 g CO₂ over the course of 12 days, which is 60 g CO₂/day per elm. Thus:

$$\frac{1187.82 \text{ kg CO}_2}{21.9 \text{ kg CO}_2/\text{year per elm}} \approx 54.2 \text{ elm years.} \quad (6.1)$$

That is, this work's estimated footprint is equivalent to roughly one elm absorbing CO₂ continuously for around 54 years, or, equivalently, about 54 elms each absorbing for one year.

6.1.2 Economic impact

In Table 4 we present the cost of each element involved in the characterization process. This includes optical components, detectors, and the student's salary, amongst others. It is worth noting that most of the hardware was present prior to my involvement, so only the newly purchased items have been included. Moreover, not all the expense information is available, so there are items missing. Hence, the total value serves as a lower-bound estimate.

Regarding the execution of the project, a mass-produced version of this source greatly benefits from the CMOS-compatible SOI platform chosen. Fabrication process would be shared with the broader semiconductor industry, yielding substantially lower costs than an equivalent TFLN or AlGaAs source. Moreover, as already mentioned in the previous section, given that the existing telecom infrastructure could be used, no additional costs would be required for the deployment of the network, beyond the chip manufacturing and due detection.

Against these advantages, however, we must recall that the entanglement-certification approach used in this work relies on SNSPDs, which are expensive devices. Yet again, SPADs would be considerably cheaper to deploy and maintain.

As for risks and limitations, most notably is that there are no mass-deployed technologies such as the one presented in this work, so there are little to none comparison points with existing technology when talking about a deployed quantum network. Consequently, any cost comparison and analysis is necessarily approximate. The viability of the project would be immediately compromised if the competing platform materials would prove to yield undoubtedly better entanglement quality; that is, if the performance gap between

Component	Qty	Unit price	Subtotal
<i>Fibre-optic components</i>			
Tapered/lensed fibre (AMS/OZ Optics) ^a	10	€90.30	€903.00
<i>Optomechanics and imaging (Thorlabs)</i>			
MDT630B/M – 3-axis NanoMax stage + controller	2	€4814.52	€9629.04
HFV002 – Tapered V-groove fibre holder	2	€161.75	€323.50
VL12X12Z – Zoom lens	1	€5854.15	€5854.15
LT101/M – Translation-stage base plate	3	€56.61	€169.83
LT1/M – Translation stage	3	€422.89	€1268.67
J750/M – Vertically adjustable lab jack	1	€389.37	€389.37
<i>Laser source</i>			
PPCL550 – Tunable 1550 nm C-band laser ^b	1	€4000.00	€4000.00
<i>Labour</i>			
Student salary	7 mo.	€1100.00/mo	€7700.00
Equipment subtotal			€22 537.56
Total (incl. labour)			€30 237.56

Table 4: Estimated cost of the experimental setup used in this thesis.

silicon and other materials should fail to close. In such case, the cost advantage argued above may not be enough to sustain SOI as the chosen platform. Moreover, the costs of maintenance/update of the network have not been covered, as since there is no existing quantum network, the processes involved in such scenarios are unbeknownst to the author of this report.

6.1.3 Social impact

On a personal level, partaking on this project has undoubtedly been a transformative experience. For starters, regarding professional growth, I have acquired skills in optics laboratories, handling of nonlinear optics processes, and characterization procedures I would’ve hardly learned anywhere else. Moreover, it has been the first time I have been part of a working team such as the Optoelectronics group for an extended period of time: working collaboratively is a paramount competence in today’s society. As for personal growth, this TFG has been carried out within an incredibly rich amalgam of different nationalities, cultures, and beliefs; which has indubitably broadened my views.

Preparing this chapter prompted me to take a closer look at the supply chains behind the devices I have engaged with these past months, providing a different perspective on the impact of the work undertaken. In particular, the very process that yields the purified silicon this work depends on, has been subject to vastly documented forced-labour concerns. These nuances are easily overlooked when comparing platforms merely on technical figures of merit.

Language throughout this thesis has been reviewed to be gender-inclusive and non-discriminatory, in line with the UPC’s Ethical Code. It is pertinent to note that ICFO takes special care to the inclusiveness and rights of all LGTB+ collectives and other minorities, having a very active and dedicated team behind.

Turning our attention to the broad implementation of this on-chip polarization-entanglement

source in a satellite-based communication network, it is harder to pin down the social impact, as it is ultimately determined by who has access to the resulting network. Let us recall that entanglement-based protocols, such as device-independent QKD, offer unconditional protection against security breaches. Consequently, it is not far-fetched to assume that the near-term beneficiaries of such a network would primarily be governments (a narrative further attested by the fact that, currently, the European Union is pushing investments in this direction). History serves as precedent that big technological breakthroughs are first deployed in military or intelligence contexts, and there is no reason to believe, at the very least within the scope of this report, that this case should be any different.

Regardless, given the large expense of having access to such an infrastructure, large institutions and telecom operators would be the next candidates for its use, rather than individual users.

On another note, it would be incomplete not to consider geopolitical notions. Currently, both chip fabrication and single-photon detection rely on a small number of highly specialized foundries and suppliers. Silicon provides an advantage in this regard as, being so integrated into global infrastructure, its handling is not overly monopolized by a single country. The same can't be said for competing platforms, such as AlGaAs, principally handled by China, who has antecedents using gallium export as a geopolitical lever. This is all to say that, when dealing with extremely technical technologies, the geopolitical disruptions at any point in the supply chain can quickly become strategic bottlenecks.

Finally, it should be pointed out that no consultation with prospective end-users nor affected communities was possible at this early research state: the social-impact discussion is purely speculative. A more grounded assessment would only be possible once the technology approaches actual deployment.

6.2 Ethical repercussions

Having assessed the environmental, economic, and social dimensions of this work, it is fitting to reflect on the ethical implications of this thesis.

Let us begin by addressing the needs this work responds to. At the most immediate level, the main driver of this work is the pursuit and expansion of scientific knowledge, specifically in the field of quantum optics. The motivation behind the development of a robust, miniaturizable quantum transmitter is the imminent threat that quantum computers pose to current cryptography protocols, receiving institutional push from bodies such as the European Union towards quantum-safe communication infrastructure.

It is pertinent to ponder on the possible consequences this TFG could have beyond the aimed for this work. The first effect concerns access: were the source to be integrated into a satellite-based QKD network, the immediate beneficiaries would be governments and large institutions able to afford such infrastructure access. As stated in the previous section, it is safe to assume the first employed context would be regarding intelligence and military communications, as the cryptographic security this technology provides would be unmatched. The second consequence concerns the material dependencies the presented design carries, and its underlying implications. The choice of integrated silicon photonics is not an ethically weightless engineering decision, at it ties any future deployment of this source to the specific supply chain, with its specific labour conditions.

It must be pointed out that, unlike fields such as medicine, quantum photonics research does not carry a profession-specific code of conduct. Nevertheless, the scientific community's established ethical and professional guidelines have been followed throughout the development of this project. Moreover, the work is in accordance to the principles of both the UPC Code of Ethics and the UPC Research Integrity Code.

Beyond the debated issues, it is worth closing with a more broad reflection on the nature of the device hither presented. It could be argued that the photon-pair source itself is agnostic to its eventual use: nothing in its components (Sagnac loop and polarization-splitter rotator) discriminates between a secure-message application and a state intelligence use-case. However, it would be naive to conclude that this technology is therefore ethically neutral, as the geopolitical, economic, and cultural context surrounding its conception are hardly so. It is therefore important to acknowledge that no research direction, including this one, exists in complete isolation –beyond a research institute’s own ethical code, the funding it requires to carry out its activities is not necessarily constrained to the same ideals. Thus, the resulting research will be, if only subtly, biased and moulded towards a specific purpose.

6.3 Sustainable Development Goals

The presented work can be related to concrete targets within the United Nations’ 2030 Agenda.

The most evident link is to SDG9 (*Industry, Innovation and Infrastructure*). Target 9.5 calls for enhancing scientific research and improving technological capacity of industrial sectors; this TFG constitutes a modest advance in entanglement-source engineering.

A second, more subtle link can be drawn to SDG 13 (*Climate Action*) through Target 13.2. Photonic links are documented to dissipate less energy per bit than their copper counterparts at high data rates, and they do not require the thermal monitoring and management that copper electronics does. However, this is currently an unrealized potential, as the device presented in this work is not yet deployed to a broad network.

Finally, for completion it is pertinent to note that not only favourable connections are present. In SDG 8 (*Decent Work and Economic Growth*), Target 8.7 calls for the eradication of forced labour. Unfortunately, the documented labour concerns around silicon refinement entail that the material this chip sits on is currently part of a supply chain that undermines this target.

6.4 Conclusions

All things considered, this analysis confirms that the source characterized in this thesis has several advantages: it builds on mature, CMOS-compatible fabrication process rather than a dedicated infrastructure, generates entanglement at telecom wavelength, and operates at room temperature, avoiding the active cooling burden of the classical electronics platforms. At the same time, we have outlined other drawbacks such as the energy cost of single-photon detection, or the social impacts of the silicon supply chain.

In conclusion, this project demonstrates that it is possible to advance toward more sustainable technologies, but also shows that authentic sustainability requires a critical analysis of impacts across all project stages.

A Automation codes for stimulated four-wave mixing data acquisition

A.1 Seed laser wavelength sweep

```
import pyvisa
import time
import csv
import os

def run_fwm_sweep():
    rm = pyvisa.ResourceManager()

    new_folder = "wvg6_TM1"
    os.makedirs(new_folder, exist_ok=True)

    santec_ip = "192.168.1.100"
    osa_ip = "192.168.55.10"

    # Seed parameters
    start_wl = 1480.0
    end_wl = 1625.0
    step_size = 1.0

    # OSA parameters
    center_wl = 1550.000
    osa_span = 160.0
    osa_res = 0.3

    try:
        print("Booting_sequence ... ")

        # ——— SANTEC SEED LASER CONFIG ———
        print("Connecting_to_Santec_Seed_Laser ... ")
        seed = rm.open_resource(f"TCPIP0::{santec_ip}::5000::SOCKET")
        seed.write_termination = "\r"
        seed.read_termination = "\r"
        seed.timeout = 5000

        seed.write("*CLS")
        seed.write(":POW:STAT_1")
        time.sleep(0.5)
        seed.write(":POW_10.00")
        seed.query("*OPC?")

        # ——— YOKOGAWA OSA CONFIG ———
        print("Connecting_to_Yokogawa_OSA ... ")
        osa = rm.open_resource(f"TCPIP0::{osa_ip}::10001::SOCKET")
        osa.write_termination = "\r\n"
        osa.read_termination = "\n"
```

```

osa.timeout = 60000

osa.query('open_anonymous')
osa.query("")

print(f"OSA:_{center_wl}_nm,{osa_span}_nm...")
osa.write(f":SENS:WAV:CENT_{center_wl}nm")
osa.write(f":SENS:WAV:SPAN_{osa_span}nm")
osa.write(f":SENS:BWID:RES_{osa_res}nm")

print(f"\n._Data_will_be_saved_in:./{new_folder}/")
current_wl = start_wl

while round(current_wl, 3) <= round(end_wl, 3):
    wl_str = f"{current_wl:.3f}"
    print(f"\n——STEP: _Seed_at_{wl_str}_nm——")

    # 1. Move seed
    seed.write(f":WAV_{wl_str}")
    seed.query("*OPC?")
    time.sleep(1.5)

    # 2. Force Trace A to accept new data
    osa.write(":TRAC:ATTR:TRA_WRIT")

    # 3. Trigger OSA
    print("Acquiring_OSA_trace...")
    osa.write(":INIT:SMOD_SING")
    osa.write(":INIT")

    time.sleep(0.5)

    osa.query("*OPC?")

    # 6. Download data
    print("Downloading_array_data...")
    raw_x = osa.query(":TRAC:DATA:X?_TRA")
    raw_y = osa.query(":TRAC:DATA:Y?_TRA")

    wls = [float(val) for val in raw_x.split(",")]
    pws = [float(val) for val in raw_y.split(",")]

    # 7. Save to CSV
    filename = f"fwm_sweep_{wl_str}.csv"
    filepath = os.path.join(new_folder, filename)

    with open(filepath, mode="w", newline="") as csv_file:
        writer = csv.writer(csv_file)
        writer.writerow(["Wvl [m]", "P [dBm]"])

```

```

        for x, y in zip(wls, pws):
            writer.writerow([x, y])

    print(f"Saved_{len(wls)}_points_to:{filepath}")
    current_wl += step_size

    print("\nWavelength_sweep_completed.")

except pyvisa.errors.VisaIOError as e:
    print(f"\nALERTA_MEDUSIL:{e}")
except Exception as e:
    print(f"\nALERTA_MEDUSIL:{e}")

finally:
    print("\nClosing_connections...")
    if "seed" in locals():
        seed.close()
    if "osa" in locals():
        try:
            osa.write("close")
            time.sleep(0.2)
        except:
            pass
    osa.close()

if __name__ == "__main__":
    run_fwm_sweep()

```

A.2 Single OSA trace acquisition

```

import pyvisa
import csv
import os
import sys

def acquire_single_trace(pump_power_dbm):
    rm = pyvisa.ResourceManager()
    osa_ip = "192.168.55.10"

    folder = "Pump_Sweep_02"
    os.makedirs(folder, exist_ok=True)

    try:
        print(f"Connecting_OSA")
        osa_rm = rm.open_resource(f"TCPIP0::{osa_ip}::10001::SOCKET")
        osa_rm.write_termination_="\r\n"
        osa_rm.read_termination_="\n"
        osa_rm.timeout_ = 10000
    except:

```

```

#####osa.query('open_"anonymous"')
#####osa.query(" ")

#####print("Downloading trace A...")
#####raw_x=osa.query(":TRAC:DATA:X? TRA")
#####raw_y=osa.query(":TRAC:DATA:Y? TRA")

#####wls=[float(val)for val in raw_x.split(",")]
#####pws=[float(val)for val in raw_y.split(",")]

#####filename=f"pump_power_sweep_{pump_power_dbm}dBm.csv"
#####filepath=os.path.join(folder, filename)

#####with open(filepath, mode="w", newline="") as csv_file:
#####    writer=csv.writer(csv_file)
#####    writer.writerow(["Wavelength [m]", "Power [dBm]"])
#####    for x, y in zip(wls, pws):
#####        writer.writerow([x, y])

#####print(f"[OK] Dades guardades a {filepath}")

#####except Exception as e:
#####    print(f"Error: {e}")
#####finally:
#####    if 'osa' in locals():
#####        osa.close()

if __name__=="__main__":
#####    if len(sys.argv)<2:
#####        print("python guarda_osa_power.py <Potencia_Pump_dBm>")
#####        print("Exemple: python guarda_osa_power.py 16.0")
#####    else:
#####        power=sys.argv[1]
#####        acquire_single_trace(power)

```

B CHSH inequality S parameter extraction

```
% =====  
%%                               ENTANGLEMENT INTERFERENCE + CHSH  
% =====  
  
clear all; close all; format long g; clc;  
  
set(groot, 'defaultTextInterpreter', 'latex');  
set(groot, 'defaultAxesTickLabelInterpreter', 'latex');  
set(groot, 'defaultLegendInterpreter', 'latex');  
  
% *****  
%%                               Data processing  
% *****  
tau = 400 * 1e-12; % coincidence window 400ps  
acq_time = 10;  
  
data_path = '../.. / Tests/timetagger/wvg2/entr9';  
files = dir(fullfile(data_path, '*A*-B*.txt'));  
  
% 1: HWP 0deg -> Alice projects to 0 deg (Ch1), 90 deg (Ch2)  
ang_B_1 = [];  
A0_B     = []; A0_B90 = [];  
A90_B    = []; A90_B90 = [];  
  
% Procedure 2: HWP at 22.5 deg -> Alice projects 45, -45  
ang_B_2 = [];  
A45_B   = []; A45_B90 = [];  
Am45_B  = []; Am45_B90 = [];  
  
for k = 1:length(files)  
    file_path = fullfile(data_path, files(k).name);  
    data = readmatrix(file_path, 'NumHeaderLines', 12);  
  
    RM_Alice = data(1, 13);  
    RM_Bob   = data(1, 12) * 2; % Bob's angle delta_B  
  
    % Single counts  
    ch1 = data(1, 3);  
    ch2 = data(1, 4);  
    ch3 = data(1, 5);  
    ch4 = data(1, 6);  
  
    % Raw Coincidences  
    raw_13 = data(1, 8); % Alice Transmitted & Bob Transmitted  
    raw_24 = data(1, 9); % Alice Reflected & Bob Reflected  
    raw_14 = data(1, 10); % Alice Transmitted & Bob Reflected  
    raw_23 = data(1, 11); % Alice Reflected & Bob Transmitted
```

```

% Accidentals calculation
acc_13 = ch1 * ch3 * tau;
acc_14 = ch1 * ch4 * tau;
acc_24 = ch2 * ch4 * tau;
acc_23 = ch2 * ch3 * tau;

if RM_Alice == 0.0 % Procedure 1
    ang_B_1 = [ang_B_1, RM_Bob];
    A0_B     = [A0_B,     raw_13 - acc_13];
    A0_B90   = [A0_B90,  raw_14 - acc_14];
    A90_B    = [A90_B,   raw_23 - acc_23];
    A90_B90  = [A90_B90, raw_24 - acc_24];
elseif RM_Alice == 22.5 % Procedure 2
    ang_B_2 = [ang_B_2, RM_Bob];
    A45_B   = [A45_B,   raw_13 - acc_13];
    A45_B90 = [A45_B90, raw_14 - acc_14];
    Am45_B  = [Am45_B,  raw_23 - acc_23];
    Am45_B90 = [Am45_B90, raw_24 - acc_24];
end
end

% Sort Procedure 1
[ang_B1_sort, idx_1] = sort(ang_B_1, 'ascend');
A0_B   = A0_B(idx_1);
A0_B90 = A0_B90(idx_1);
A90_B  = A90_B(idx_1);
A90_B90 = A90_B90(idx_1);

% Sort Procedure 2
[ang_B2_sort, idx_2] = sort(ang_B_2, 'ascend');
A45_B   = A45_B(idx_2);
A45_B90 = A45_B90(idx_2);
Am45_B  = Am45_B(idx_2);
Am45_B90 = Am45_B90(idx_2);

% Poisson Error Calculation: sqrt(True_Counts) / T
err_A0_B   = sqrt(max(A0_B, 0) * acq_time) / acq_time;
err_A0_B90 = sqrt(max(A0_B90, 0) * acq_time) / acq_time;
err_A90_B  = sqrt(max(A90_B, 0) * acq_time) / acq_time;
err_A90_B90 = sqrt(max(A90_B90, 0) * acq_time) / acq_time;

err_A45_B   = sqrt(max(A45_B, 0) * acq_time) / acq_time;
err_A45_B90 = sqrt(max(A45_B90, 0) * acq_time) / acq_time;
err_Am45_B  = sqrt(max(Am45_B, 0) * acq_time) / acq_time;
err_Am45_B90 = sqrt(max(Am45_B90, 0) * acq_time) / acq_time;

% *****
%%      Cosine fit & Visibility
% *****

```

```

cosineModel = @(p, x) p(1) * cos(p(2) * x + p(3)) + p(4);
opts = optimoptions('lsqcurvefit', 'Display', 'off');
fitCosine = @(x, y) local_fit_cosine(x, y, cosineModel, opts);

% Fit Procedure 1
[p_A0_B, xq_1, yq_A0_B] = fitCosine(ang_B1_sort, A0_B);
[p_A0_B90,~, yq_A0_B90] = fitCosine(ang_B1_sort, A0_B90);
[p_A90_B, ~, yq_A90_B] = fitCosine(ang_B1_sort, A90_B);
[p_A90_B90,~, yq_A90_B90]= fitCosine(ang_B1_sort, A90_B90);

% Fit Procedure 2
[p_A45_B, xq_2, yq_A45_B] = fitCosine(ang_B2_sort, A45_B);
[p_A45_B90,~, yq_A45_B90] = fitCosine(ang_B2_sort, A45_B90);
[p_Am45_B, ~, yq_Am45_B] = fitCosine(ang_B2_sort, Am45_B);
[p_Am45_B90,~, yq_Am45_B90]= fitCosine(ang_B2_sort, Am45_B90);

% Visibilities:  $V = |a| / d$ 
v_A0_B = abs(p_A0_B(1)) / p_A0_B(4);
v_A0_B90 = abs(p_A0_B90(1)) / p_A0_B90(4);
v_A90_B = abs(p_A90_B(1)) / p_A90_B(4);
v_A90_B90= abs(p_A90_B90(1))/ p_A90_B90(4);

v_A45_B = abs(p_A45_B(1)) / p_A45_B(4);
v_A45_B90 = abs(p_A45_B90(1)) / p_A45_B90(4);
v_Am45_B = abs(p_Am45_B(1)) / p_Am45_B(4);
v_Am45_B90= abs(p_Am45_B90(1))/ p_Am45_B90(4);

overall_vis_0_90 = mean([v_A0_B, v_A0_B90, v_A90_B, v_A90_B90]);
overall_vis_45_m45 = mean([v_A45_B, v_A45_B90, v_Am45_B, v_Am45_B90]);

fprintf('——_INTERFERENCE_VISIBILITIES_——\n');
fprintf('Overall_Rectilinear_Basis_Visibility:_%%.2f%%\n', ...
    overall_vis_0_90 * 100);
fprintf('Overall_Diagonal_Basis_Visibility:_%%.2f%%\n', ...
    overall_vis_45_m45 * 100);

% *****
%% CHSH: three angle choices, side by side
% *****

% Normalized joint correlators  $E(\theta_A, \theta_B)$ 
% Proc 1 → Alice fixed at 0 deg; Proc 2 → Alice fixed at 45 deg
E1_raw = (A0_B + A90_B90 - A0_B90 - A90_B) ./ ...
    (A0_B + A90_B90 + A0_B90 + A90_B); %  $E(0, \theta_B)$ 
E2_raw = (A45_B + Am45_B90 - A45_B90 - Am45_B) ./ ...
    (A45_B + Am45_B90 + A45_B90 + Am45_B); %  $E(45, \theta_B)$ 

% Fit each correlator vs Bob's angle so we can evaluate  $E$  at ANY angle,

```

```

% not just the ones actually measured
[pE1, ~, ~] = fitCosine(ang_B1_sort, E1_raw);
[pE2, ~, ~] = fitCosine(ang_B2_sort, E2_raw);
E1fun = @(th) cosineModel(pE1, th); % E(0, theta_B), continuous
E2fun = @(th) cosineModel(pE2, th); % E(45, theta_B), continuous

% % — (1) Literal supervisor angles: Bob at 0 / 45 —
% [S1, sS1] = chsh_at(0, 45, ang_B1_sort, ang_B2_sort, ...
%     A0_B, A0_B90, A90_B, A90_B90, ...
%     A45_B, A45_B90, Am45_B, Am45_B90, ...
%     err_A0_B, err_A0_B90, err_A90_B, err_A90_B90, ...
%     err_A45_B, err_A45_B90, err_Am45_B, err_Am45_B90, E1fun, E2fun);

% — (2) Textbook optimal (assumes exactly 45 deg phase mismatch): —
% Bob at 22.5 / 67.5
[S2, sS2] = chsh_at(22.5, 67.5, ang_B1_sort, ang_B2_sort, ...
    A0_B, A0_B90, A90_B, A90_B90, ...
    A45_B, A45_B90, Am45_B, Am45_B90, ...
    err_A0_B, err_A0_B90, err_A90_B, err_A90_B90, ...
    err_A45_B, err_A45_B90, err_Am45_B, err_Am45_B90, E1fun, E2fun);

% — (3) True optimal from grid search over the fitted curves —
% (accounts for the real ~55 deg phase mismatch measured on this hardware)
bgrid = 0:0.05:360;
[B, Bp] = meshgrid(bgrid, bgrid);
% NOTE: meshgrid(bgrid, bgrid) gives B(i,j)=bgrid(j), Bp(i,j)=bgrid(i) —
% i.e. the FIRST index (i) corresponds to "bp" and the SECOND (j) to "b".
% Getting this backwards silently produces a different (wrong) S, since
% S is not symmetric under swapping b <-> bp.
Sgrid = E1fun(B) - E1fun(Bp) + E2fun(B) + E2fun(Bp);
[Sgrid_best, idxMax] = max(abs(Sgrid(:)));
[i, j] = ind2sub(size(Sgrid), idxMax);
b_opt = bgrid(j); % B(i,j) = bgrid(j)
bp_opt = bgrid(i); % Bp(i,j) = bgrid(i)

[S3, sS3] = chsh_at(b_opt, bp_opt, ang_B1_sort, ang_B2_sort, ...
    A0_B, A0_B90, A90_B, A90_B90, ...
    A45_B, A45_B90, Am45_B, Am45_B90, ...
    err_A0_B, err_A0_B90, err_A90_B, err_A90_B90, ...
    err_A45_B, err_A45_B90, err_Am45_B, err_Am45_B90, E1fun, E2fun);

% Self-check: chsh_at's recomputed S must match the grid value at (i,j),
% otherwise the index mapping above is wrong again
assert(abs(abs(S3) - Sgrid_best) < 1e-9, ...
    ['Grid-search_index_mapping_mismatch:_S3_does_not_match_' ...
    'Sgrid(idxMax)._Check_meshgrid_orientation.']);

% — Also compute S directly from the raw counts at the NEAREST measured
% Bob angle (supervisor's "method 1"), as opposed to S1/S2/S3 above which

```

```

% use the smooth cosine fit ("method 2"). —
S_raw_at = @(b, bp) local_S_from_raw(b, bp, ang_B1_sort, ang_B2_sort, ...
    A0_B, A0_B90, A90_B, A90_B90, A45_B, A45_B90, Am45_B, Am45_B90);

S1_raw = S_raw_at(0, 45);
S2_raw = S_raw_at(22.5, 67.5);
S3_raw = S_raw_at(b_opt, bp_opt);

fprintf('\n——CHSH: comparison of angle choices——\n');
fprintf('%-32s%-10s%-10s%-10s%-10s%-10s\n', ...
    'Setting', 'S_fit', 'S_raw', 'sigma_S', 'N_sigma(fit)', 'N_sigma(raw)');
%fprintf('%-32s %10.4f %10.4f %10.4f %10.2f %10.2f\n', ...
%    'Literal 0/45', S1, S1_raw, sS1, (abs(S1)-2)/sS1, (abs(S1_raw)-2)/sS1);
fprintf('%-32s%-10.4f%-10.4f%-10.4f%-10.2f%-10.2f\n', ...
    'Textbook 22.5/67.5', S2, S2_raw, sS2, ...
    (abs(S2)-2)/sS2, (abs(S2_raw)-2)/sS2);
fprintf('%-32s%-10.4f%-10.4f%-10.4f%-10.2f%-10.2f\n', ...
    sprintf('True_optimal_%.1f/%.1f', b_opt, bp_opt), S3, S3_raw, sS3, ...
    (abs(S3)-2)/sS3, (abs(S3_raw)-2)/sS3);
fprintf(['\n(S_fit: correlators evaluated on the smooth cosine fit.) ...
    'S_raw: correlators\n']);
fprintf(['_taken_from_the_raw_coincidence_counts_nearest_each_target_ ...
    'angle.\n']);
fprintf(['_sigma_S_is_the_Poisson-propagated_uncertainty_from_the_raw_ ...
    'counts, used for both.)\n']);

% =====
%%                               Exact phase offset from the fits (no eyeballing)
% =====

w1 = pE1(2); phi1 = pE1(3);
w2 = pE2(2); phi2 = pE2(3);
period1 = 2*pi/w1; period2 = 2*pi/w2;    % should both be ~180 deg

theta_peak_1 = mod(-phi1/w1, period1);    % Alice=H: expect peak near Bob=0
theta_peak_2 = mod(-phi2/w2, period2);    % Alice=D: expect peak near Bob=45

raw_offset = theta_peak_2 - theta_peak_1;
% wrap near 45
measured_offset_deg = raw_offset - 180*round((raw_offset - 45)/180);

fprintf('\n——Measured_phase_offset(exact)——\n');
fprintf('Peak_of_E(0,theta_B)_at_%.2f_deg_(period_%.2f_deg)\n', ...
    theta_peak_1, period1);
fprintf('Peak_of_E(45,theta_B)_at_%.2f_deg_(period_%.2f_deg)\n', ...
    theta_peak_2, period2);
fprintf('Measured_offset: %.2f_deg_(theory_expects_exactly_45_deg)\n', ...
    measured_offset_deg);
fprintf('Excess_beyond_theory: %.2f_deg\n', measured_offset_deg - 45);

```

```

% Independent cross-check using the RAW fringe fits (not the E-correlator
% fits), for consistency
w1b=p_A0_B(2); phi1b=p_A0_B(3); w2b=p_A45_B(2); phi2b=p_A45_B(3);
theta_peak_1b = mod(-phi1b/w1b, 2*pi/w1b);
theta_peak_2b = mod(-phi2b/w2b, 2*pi/w2b);
offset_b = theta_peak_2b - theta_peak_1b;
measured_offset_deg_b = offset_b - 180*round((offset_b - 45)/180);
fprintf('Cross-check_from_raw_A0_B/A45_B_fits: %.2f deg\n', ...
    measured_offset_deg_b);

%% — Constrained bisector search: b, b' forced exactly 45 deg apart —
betaGrid = 0:0.02:180;
S_of_beta = E1fun(betaGrid-22.5) - E1fun(betaGrid+22.5) ...
    + E2fun(betaGrid-22.5) + E2fun(betaGrid+22.5);
[~, iBeta] = max(abs(S_of_beta));
beta_opt = betaGrid(iBeta);
b_con = beta_opt - 22.5;
bp_con = beta_opt + 22.5;

fprintf(' \n — Constrained_bisector_search — \n ');
fprintf(' Optimal_bisector = %.2f deg > b = %.2f, bp = %.2f\n', ...
    beta_opt, b_con, bp_con);
fprintf(' Theory-predicted_bisector (22.5 + offset/2) = %.2f deg\n', ...
    22.5 + measured_offset_deg/2);

[S_con, sS_con] = chsh_at(b_con, bp_con, ang_B1_sort, ang_B2_sort, ...
    A0_B, A0_B90, A90_B, A90_B90, ...
    A45_B, A45_B90, Am45_B, Am45_B90, ...
    err_A0_B, err_A0_B90, err_A90_B, err_A90_B90, ...
    err_A45_B, err_A45_B90, err_Am45_B, err_Am45_B90, E1fun, E2fun);
S_con_raw = S_raw_at(b_con, bp_con);

fprintf(' S_(fit) = %.4f S_(raw) = %.4f sigma_S = %.4f N_sigma(raw) = %.2f\n', ...
    S_con, S_con_raw, sS_con, (abs(S_con_raw)-2)/sS_con);

% =====
% CHSH UNCERTAINTY: weighted fit + delta-method propagation
% =====

%% — Per-point uncertainty on the correlators themselves —
err_E1_raw = arrayfun(@(i) corr_error(A0_B(i), A0_B90(i), A90_B(i), ...
    A90_B90(i), err_A0_B(i), err_A0_B90(i), err_A90_B(i), ...
    err_A90_B90(i)), 1:length(A0_B));
err_E2_raw = arrayfun(@(i) corr_error(A45_B(i), A45_B90(i), Am45_B(i), ...
    Am45_B90(i), err_A45_B(i), err_A45_B90(i), err_Am45_B(i), ...
    err_Am45_B90(i)), 1:length(A45_B));

```

```

%% — Weighted cosine fits, with parameter covariance —
[pE1_w, covp1, chi2red1] = local_fit_cosine_weighted(ang_B1_sort, ...
    E1_raw, err_E1_raw, cosineModel, opts);
[pE2_w, covp2, chi2red2] = local_fit_cosine_weighted(ang_B2_sort, ...
    E2_raw, err_E2_raw, cosineModel, opts);

fprintf('\n—Weighted-fit_diagnostics—\n');
fprintf(['E1(theta_B)_fit: reduced_chi^2= %.3f ' ...
    '(should_be ~1 if_errors+model_are_good)\n'], chi2red1);
fprintf('E2(theta_B)_fit: reduced_chi^2= %.3f\n', chi2red2);
if chi2red1 > 2 || chi2red2 > 2
    warning(['Reduced_chi^2 is well above 1—either the error bars are ' ...
        'underestimated, or the single-cosine model is not fully ' ...
        'capturing the data. Investigate before trusting sigma_S.']);
end

%% — Rigorous CHSH evaluation at any angle pair —
fprintf('\n—CHSH with rigorous (covariance-propagated) uncertainty—\n');
fprintf('%-32s %-10s %-10s %-10s\n', 'Setting', 'S', 'sigma_S', 'N_sigma');

[S_txt, sS_txt] = chsh_with_covariance(22.5, 67.5, pE1_w, covp1, ...
    pE2_w, covp2);
fprintf('%-32s %-10.4f %-10.4f %-10.2f\n', 'Textbook_22.5/67.5', S_txt, ...
    sS_txt, (abs(S_txt)-2)/sS_txt);

[S_con_rig, sS_con_rig] = chsh_with_covariance(b_con, bp_con, pE1_w, ...
    covp1, pE2_w, covp2);
fprintf('%-32s %-10.4f %-10.4f %-10.2f\n', ...
    sprintf('Constrained_bisector_(%.1f/%.1f)', b_con, bp_con), ...
    S_con_rig, sS_con_rig, (abs(S_con_rig)-2)/sS_con_rig);

[S_free_rig, sS_free_rig] = chsh_with_covariance(b_opt, bp_opt, pE1_w, ...
    covp1, pE2_w, covp2);
fprintf('%-32s %-10.4f %-10.4f %-10.2f\n', ...
    sprintf('Free_optimum_(%.1f/%.1f)', b_opt, bp_opt), ...
    S_free_rig, sS_free_rig, (abs(S_free_rig)-2)/sS_free_rig);

fprintf('\nConsistency_check: constrained_vs_free_optimum_S_differ_by %.4f\n',
    abs(abs(S_con_rig) - abs(S_free_rig)));
fprintf(['(Small_difference_here_confirms_the_45-degree_assumption ' ...
    'cost_is_negligible.)\n']);

fprintf('\n—HEADLINE_RESULT_FOR_THESIS—\n');
fprintf('S= %.4f +/- %.4f (0.2f_sigma_above_the_classical_bound_of_2)\n', ...
    S_con_rig, sS_con_rig, (abs(S_con_rig)-2)/sS_con_rig);

%% =====
% LOCAL FUNCTIONS

```

%

```
function [pFit, xq, yq] = local_fit_cosine(x, y, cosineModel, opts)
    x = x(:); y = y(:);
    d0 = mean(y);
    a0 = (max(y) - min(y)) / 2;
    b0 = 2*pi/180;
    [~, idx_max] = max(y);
    c0 = -b0 * x(idx_max);
    p0 = [a0, b0, c0, d0];

    lb = [-Inf, 0.5*b0, -Inf, -Inf];
    ub = [ Inf, 1.5*b0,  Inf,  Inf];
    pFit = lsqcurvefit(cosineModel, p0, x, y, lb, ub, opts);

    xq = linspace(min(x), max(x), 300);
    yq = cosineModel(pFit, xq);
```

end

```
function [S, sigma_S, ib, ibp] = chsh_at(b, bp, ang1, ang2, ...
    A0_B, A0_B90, A90_B, A90_B90, ...
    A45_B, A45_B90, Am45_B, Am45_B90, ...
    err_A0_B, err_A0_B90, err_A90_B, err_A90_B90, ...
    err_A45_B, err_A45_B90, err_Am45_B, err_Am45_B90, ...
    E1fun, E2fun)
    % Computes CHSH S at settings (b, bp) using the fitted E-functions,
    % and propagates Poissonian error using the raw coincidence counts
    % nearest to those angles.

    [~, ib1] = min(abs(ang1 - b));
    [~, ib1p] = min(abs(ang1 - bp));
    [~, ib2] = min(abs(ang2 - b));
    [~, ib2p] = min(abs(ang2 - bp));

    S = E1fun(b) - E1fun(bp) + E2fun(b) + E2fun(bp);

    sig1 = corr_error(A0_B(ib1), A0_B90(ib1), A90_B(ib1), A90_B90(ib1), ...
        err_A0_B(ib1), err_A0_B90(ib1), err_A90_B(ib1), ...
        err_A90_B90(ib1));
    sig1p = corr_error(A0_B(ib1p), A0_B90(ib1p), A90_B(ib1p), A90_B90(ib1p), ...
        err_A0_B(ib1p), err_A0_B90(ib1p), err_A90_B(ib1p), ...
        err_A90_B90(ib1p));
    sig2 = corr_error(A45_B(ib2), A45_B90(ib2), Am45_B(ib2), Am45_B90(ib2), ...
        err_A45_B(ib2), err_A45_B90(ib2), err_Am45_B(ib2), ...
        err_Am45_B90(ib2));
    sig2p = corr_error(A45_B(ib2p), A45_B90(ib2p), Am45_B(ib2p), ...
        Am45_B90(ib2p), err_A45_B(ib2p), err_A45_B90(ib2p), ...
        err_Am45_B(ib2p), err_Am45_B90(ib2p));
```

```

    % Combined in quadrature: valid because these four correlators are
    % measured at four different (independent) Bob settings / photon samples
    sigma_S = sqrt(sig1^2 + sig1p^2 + sig2^2 + sig2p^2);
    ib = b; ibp = bp;
end

function S = local_S_from_raw(b, bp, ang1, ang2, A0_B, A0_B90, A90_B, ...
    A90_B90, A45_B, A45_B90, Am45_B, Am45_B90)
    % CHSH S computed directly from the raw coincidence counts nearest to
    % the requested Bob angles b, bp (no fitting involved).
    [~, ib1] = min(abs(ang1 - b));
    [~, ib1p] = min(abs(ang1 - bp));
    [~, ib2] = min(abs(ang2 - b));
    [~, ib2p] = min(abs(ang2 - bp));

    E_a0_b = (A0_B(ib1) + A90_B90(ib1) - A0_B90(ib1) - A90_B(ib1))
    / ...
        (A0_B(ib1) + A90_B90(ib1) + A0_B90(ib1) + A90_B(ib1));
    E_a0_bp = (A0_B(ib1p) + A90_B90(ib1p) - A0_B90(ib1p) - A90_B(ib1p)) / ...
        (A0_B(ib1p) + A90_B90(ib1p) + A0_B90(ib1p) + A90_B(ib1p));
    E_a45_b = (A45_B(ib2) + Am45_B90(ib2) - A45_B90(ib2) - Am45_B(ib2))
    / ...
        (A45_B(ib2) + Am45_B90(ib2) + A45_B90(ib2) + Am45_B(ib2));
    E_a45_bp = (A45_B(ib2p) + Am45_B90(ib2p) - A45_B90(ib2p) - Am45_B(ib2p)) /
        (A45_B(ib2p) + Am45_B90(ib2p) + A45_B90(ib2p) + Am45_B(ib2p));

    S = E_a0_b - E_a0_bp + E_a45_b + E_a45_bp;
end

function sigE = corr_error(a, b, c, d, sa, sb, sc, sd)
    % Poissonian error propagation for  $E = (a+d-b-c)/(a+b+c+d)$ 
    % a,b,c,d are independent coincidence counts with std devs sa,sb,sc,sd
    N = a + b + c + d;
    dEda = 2 * (b + c) / N^2; % = dE/dd (a and d enter symmetrically)
    dEdb = -2 * (a + d) / N^2; % = dE/dc (b and c enter symmetrically)
    sigE = sqrt( dEda^2 * (sa^2 + sd^2) + dEdb^2 * (sb^2 + sc^2) );
end

function [pFit, covp, chi2red, xq, yq] = local_fit_cosine_weighted(...
    x, y, sy, cosineModel, opts)
    % Weighted nonlinear least-squares fit of the cosine model, using known
    % per-point standard deviations sy, returning the parameter covariance
    % matrix (for delta-method error propagation) and reduced chi-square
    % (diagnostic: should be ~1 if error bars and model are both trustworthy).

    x = x(:); y = y(:); sy = sy(:);
    sy(sy <= 0 | isnan(sy)) = median(sy(sy > 0)); % guard degenerate errors

    d0 = mean(y);

```

```

a0 = (max(y) - min(y)) / 2;
b0 = 2*pi/180;
[~, idx_max] = max(y);
c0 = -b0 * x(idx_max);
p0 = [a0, b0, c0, d0];

lb = [-Inf, 0.5*b0, -Inf, -Inf];
ub = [ Inf, 1.5*b0, Inf, Inf];

% Fit in WEIGHTED (chi-square) units: residual = (model - data)/sigma
wModel = @(p, xx) cosineModel(p, xx) ./ sy;
yWeighted = y ./ sy;

[pFit, resnorm, residual, ~, ~, ~, J] = ...
    lsqcurvefit(wModel, p0, x, yWeighted, lb, ub, opts);

J = full(J);
dof = length(y) - length(pFit);
chi2red = resnorm / dof;

% Since residuals are already in units of sigma, (J'J)^-1 directly
% gives the parameter covariance (no extra MSE rescaling needed).
covp = inv(J' * J); %ok<MINV>

xq = linspace(min(x), max(x), 300);
yq = cosineModel(pFit, xq);
end

function [Eval, sigmaE, grad] = evalCosineWithUnc(pFit, covp, theta)
% Evaluates the cosine model a*cos(w*theta+c)+d at a given theta, along
% with its delta-method standard error, using the fit's parameter
% covariance matrix covp.

a = pFit(1); w = pFit(2); c = pFit(3); d = pFit(4);
Eval = a*cos(w*theta + c) + d;

% Analytic gradient wrt [a, w, c, d]
grad = [ cos(w*theta + c); ...
        -a*theta*sin(w*theta + c); ...
        -a*sin(w*theta + c); ...
        1 ];

sigmaE = sqrt(grad' * covp * grad);
end

function [S, sigma_S] = chsh_with_covariance(b, bp, pE1, covp1, pE2, covp2)
% Rigorous CHSH S and its uncertainty at arbitrary (possibly unmeasured)
% angles b, bp, via delta-method propagation of the weighted-fit
% covariance matrices of E1 and E2. E1, E2 come from statistically

```

*% independent measurement runs (Procedure 1 vs Procedure 2), so their
 % contributions to sigma_S add in quadrature; WITHIN each fit, the
 % correlation between evaluations at b and bp (both derived from the
 % same 4 fit parameters) is correctly captured via the shared gradient.*

```
[E1b, ~, g1b] = evalCosineWithUnc(pE1, covp1, b);
[E1bp, ~, g1bp] = evalCosineWithUnc(pE1, covp1, bp);
[E2b, ~, g2b] = evalCosineWithUnc(pE2, covp2, b);
[E2bp, ~, g2bp] = evalCosineWithUnc(pE2, covp2, bp);
```

```
S = E1b - E1bp + E2b + E2bp;
```

```
v = g1b - g1bp;    % gradient of (E1(b) - E1(b')) wrt pE1
u = g2b + g2bp;    % gradient of (E2(b) + E2(b')) wrt pE2
```

```
sigma_S = sqrt(v' * covp1 * v + u' * covp2 * u);
```

```
end
```

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